



AVEVA Licensing System
v3.0.0.0

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Revision Sheet

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AVEVA Licensing System

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1 Introduction

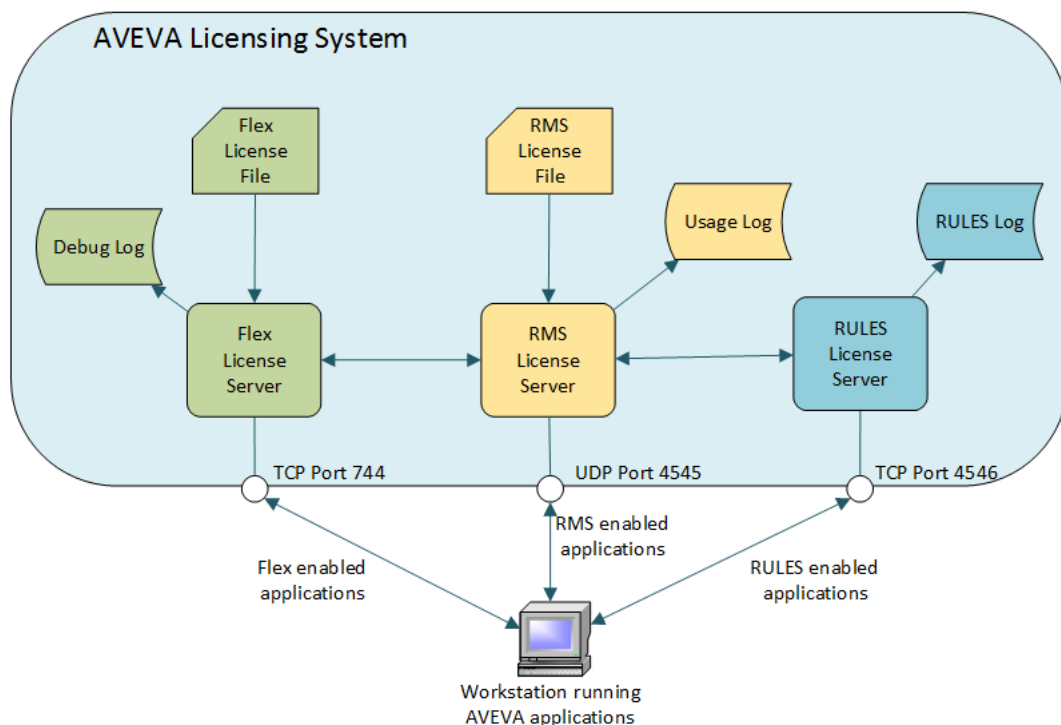
The AVEVA Licensing System (ALS) consists of three license servers:

1. A Flex license server provided by Flexera.
2. A Rights Management System (RMS) license server provided by Gemalto.
3. A Reserved User Licensing Enforcement System (RULES) license server proprietary to AVEVA.

All three license servers work in a cooperative manner that effectively allows the delivery of licenses from a single "pool" of licenses. This means that different applications, that are licensed with different technologies, are able to share those license features. This is of particular benefit when customers use different versions of the same application, some of which are licensed with RMS, and some licensed with Flex. Products can be licensed without the need to purchase duplicate sets of licenses for each license server.

In general, new AVEVA products are licensed via the RULES server, more recent AVEVA products are licensed with RMS, and older products are licensed with Flex.

The following diagram shows a typical ALS configuration using the default license server ports:



The ALS consists of the following components:

RMS License Server:	Services license requests from RMS-enabled applications.
RMS License File:	Contains the licenses that can be served by the RMS License Server and the RULES license server.
Usage Log:	A log file written by the RMS License Server that contains details of license check-out and check-in operations performed by RMS-enabled applications.
Flex License Server:	Services license requests from Flex-enabled applications.
Flex License File:	Contains the licenses that can be served by the Flex License Server.
Debug Log:	A log file written by the Flex License Server that contains details of license check-out and check-in operations performed by Flex-enabled applications.
RULES License Server	Services license requests for RULES-enabled applications. RULES is also a proxy for all RMS licenses. Refer to RULES SERVER for further information.
RULES Log	A log file written by the RULES server.

There are three ports configured in the ALS, one for each license server.

The ALS uses Dual License Server technology to allow applications that use the different RMS and Flex licensing technologies to share license entitlement. This has been achieved by configuring the Flex License Server so that it notifies the RMS License Server with details of the licensing activities of the Flex-enabled applications as and when they occur.

The RMS License Server effectively "controls" the Flex License Server and its clients by allowing or disallowing Flex license check-out and check-in operations based on the licenses present in the RMS license file.

The RULES license server acts as a proxy for RMS licensing, and enforces licensing for "Named Users". Refer to [RULES SERVER](#) for more information.

Log Files

The contents of the RMS Usage Log file, the Flex Debug Log file and the RULES log file reflect the operations of each license server and the use of licenses by each kind of AVEVA application.

Important: The DLS functionality means that the RMS Usage Log also contains the details of the Flex licensing operations.

ALS Windows Services

The three ALS license servers are configured to run as Windows Services. They can be identified by the following names in the Windows Services panel:

RMS License Server:	Sentinel RMS License Manager_AVEVA
Flex License Server:	AVEVA DLS Flex Server
RULES License Server:	AVEVA RULES Server

AVEVA License System Console

To make sure that the ALS can be correctly and easily administered, an [AVEVA License System Console](#) has been provided that grants easy access to the administrative tasks that most customers are likely to undertake and presents the underlying licensing systems in a unified fashion. The ALS Console provides easy access to the following functionality:

- Starting/Stopping the ALS
- License File Installation
- ALS Server Status Query

Client Configuration Tool

The [ALS Client Configuration Tool](#) has been provided so that client PCs can be easily configured with details of up to three AVEVA Licensing Systems.

The Client configuration tool provides easy access to the Server Port Configuration.

Note: The ALS has been specifically designed so that existing FlexMan clients do not need to be reconfigured if the ALS Server is installed using the default port settings.

2 AVEVA License System Console

The AVEVA License System Console, or ALS Console, is a tool that allows the system administrator to manage AVEVA licenses. The system administrator is not to be confused with the AVEVA License Administrator (who supplies licenses) or the end user (who consumes licenses).

If the system administrator is unfamiliar with the ALS, an explanation is provided. Refer to [Introduction](#) for further information.

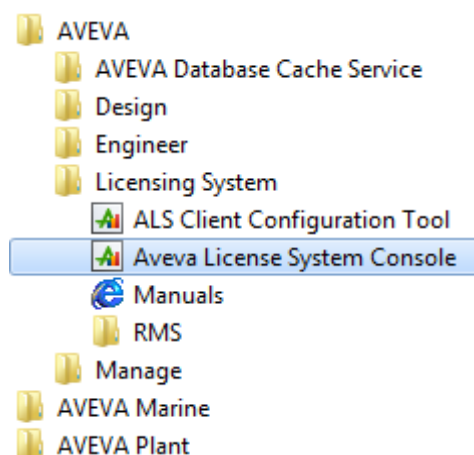
The ALS consists of three separate license servers that run cooperatively. Due to the complexity of configuring the three separate license servers so that they can operate together, a dedicated utility has been provided called the AVEVA License System Console.

The AVEVA License System Console is an easy-to-use general user interface that can be used to perform tasks such as license file installation and stopping/starting the license system.

While it is possible to manually configure the ALS by directly editing configuration files, AVEVA strongly recommends that only the AVEVA License System Console is used to administer the ALS.

The AVEVA License System Console is installed as part of the ALS installation. Refer to *AVEVA Licensing System 3.0 Installation Guide* for further information.

To launch the **AVEVA License System Console** select **Start > All Programs > AVEVA > Licensing System > AVEVA License System Console**.



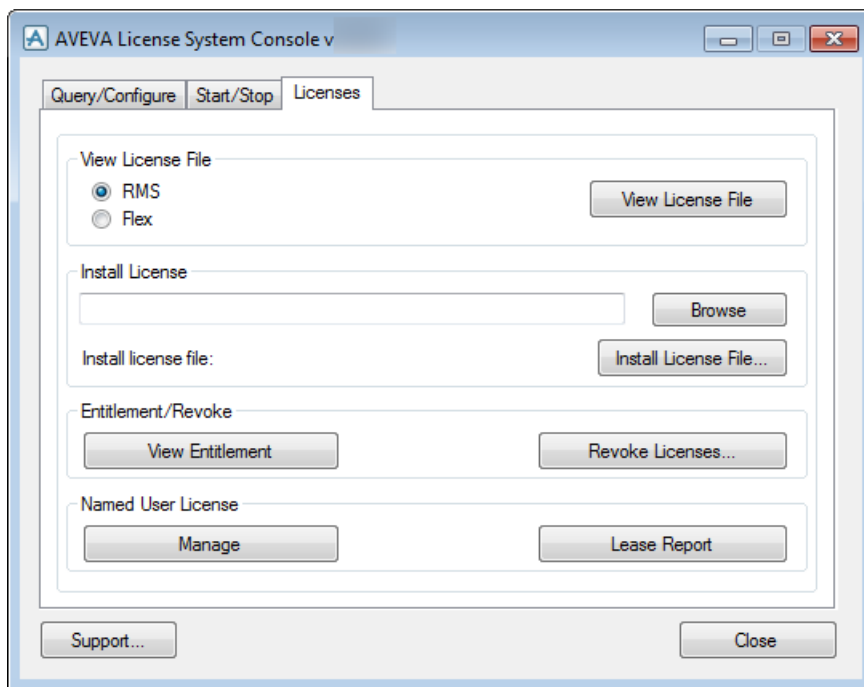
The AVEVA License System Console consists of three tabs:

- [Licenses](#): allows you to install license files and product entitlements, view individual license files.

- [Start/Stop](#): allows you to start and stop the license servers.
- [Query/Configure](#): allows you to query the server status, view log files and edit the license server configuration.

2.1 Licenses

The **Licenses** tab displays the following window:



View License File

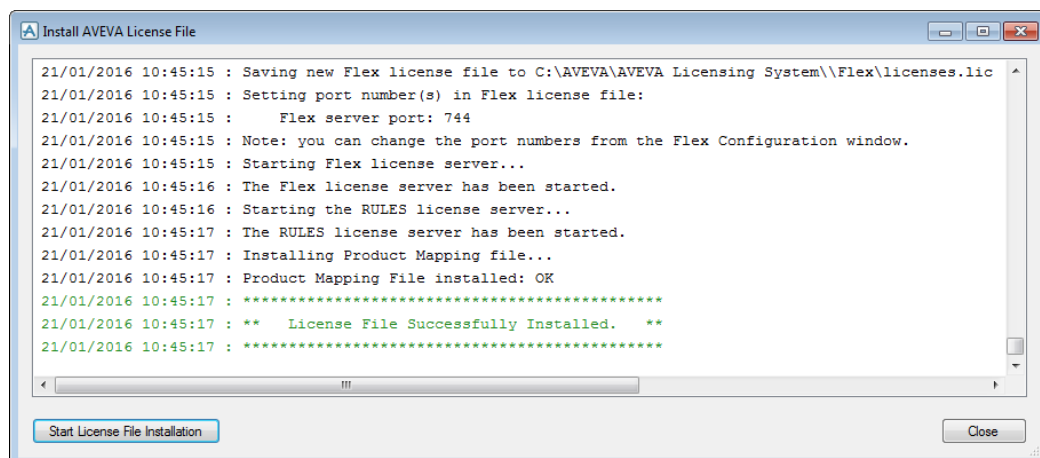
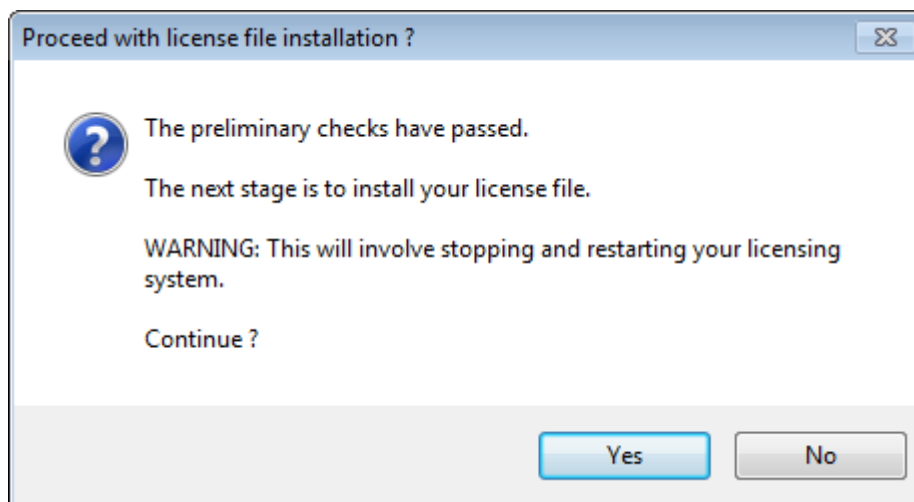
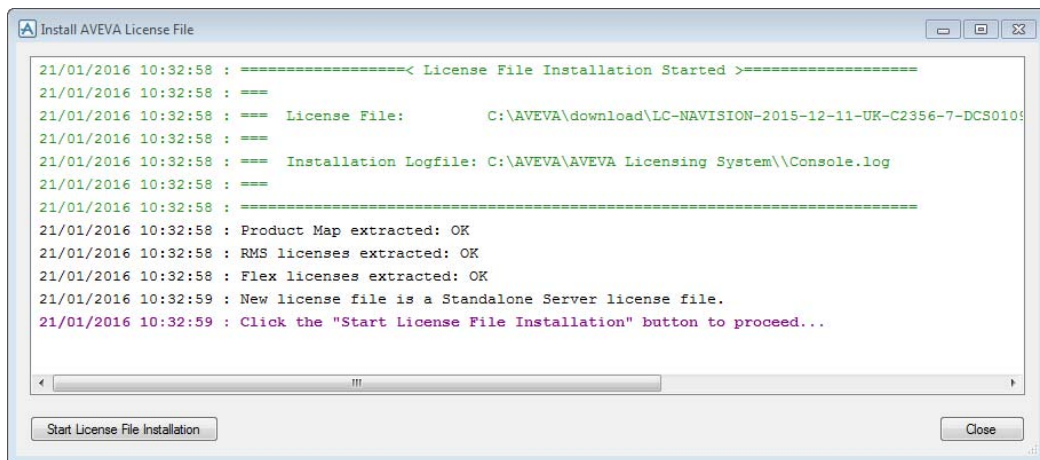
Use the radio button to select the required license file type and click **View License File** to display the appropriate license file.

Install License

Note: ALS uses clock tampering detection mechanisms, changing the system clock may cause the system to fail.

1. Click **Browse** to locate the license file.
2. Click **Install License File**.
3. Click **Start License File Installation**.
4. Click **Yes** in the **Warning** message window.

The licenses are installed, the license servers are restarted and a successful message is displayed.

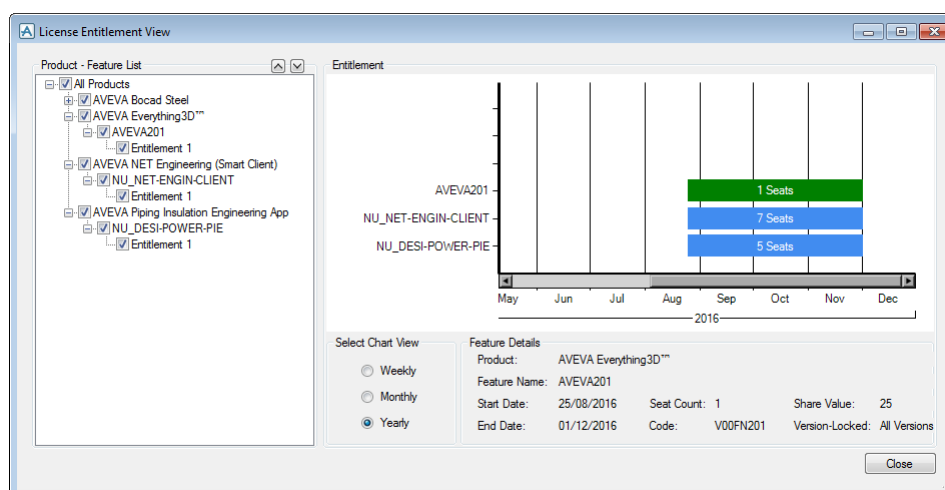


Note: Backups of existing license files are created.

Note: A check and install log of the license file installation is written to a Console.log file. The log file is stored in the default folder. For example, C:\AVEVA\AVEVA Licensing System. If there are any issues when installing the license file, send the Console.log file to your regional support team.

View Entitlement

Clicking **View Entitlement** opens a new window that allows you to view the product licenses. The window displays the number of licenses available for each product and the license expiry date.

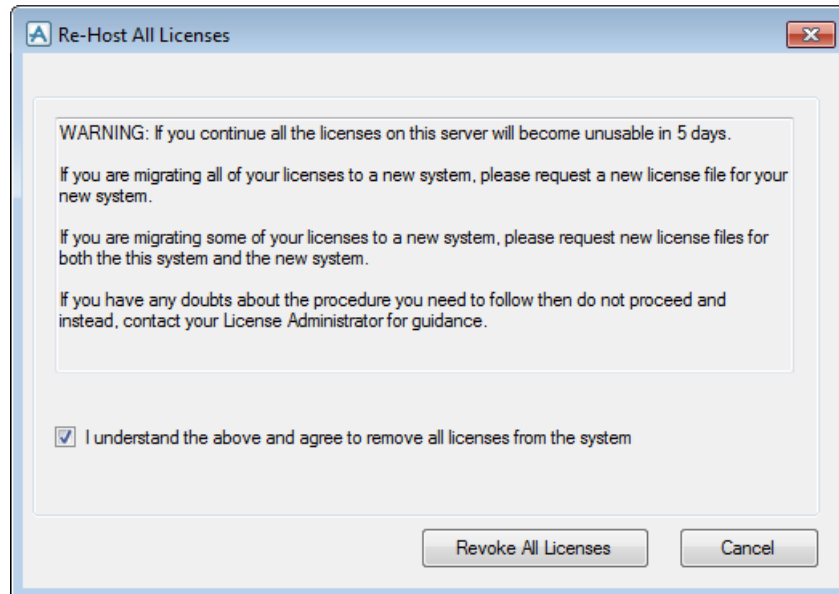


Operations

1. The product mapping file is located at the following location:
%AVEVA_LICENSING%\ProductMaps
2. The list of products in the product mapping file are listed on the left screen in a tree format. The display of the tree format can be expanded or collapsed by clicking the  buttons.
3. When a node is expanded for a particular product, all the features belonging to that product are listed.
4. When a feature is expanded, the licenses for that feature are listed. For example, in the above screenshot, the AVEVA Everything3D™ product contains one license (Entitlement 1) consisting of one seat.
5. When a specific feature is selected, via a check box, the entitlements for that feature are displayed on the right side of the window in the entitlement chart.
6. The date range on the X axis of the entitlement chart displays the license start date and end date. The date range can be displayed weekly, monthly or yearly.
7. When a specific feature is selected on the entitlement chart, product details for that feature are displayed on the bottom-right of the window.

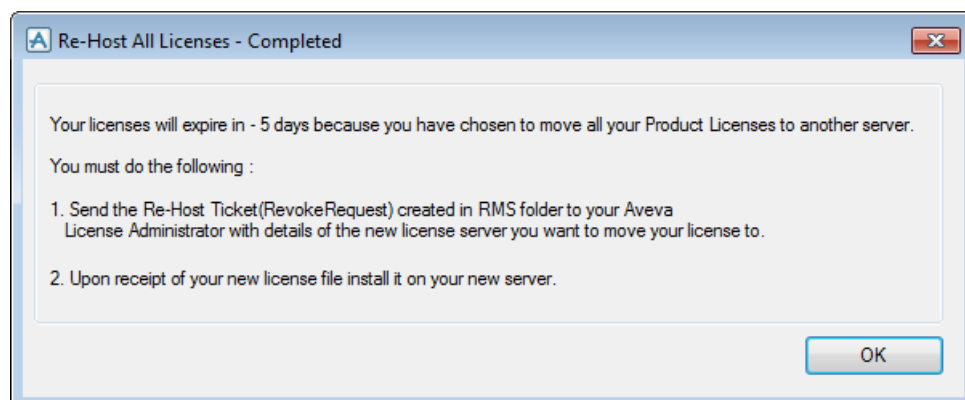
Revoke Licenses

Clicking **Revoke Licenses** opens a new window that provides a description of the revoke licenses processes.



1. Once you have read the description and selected the **I understand the above and agree to remove all licenses from the system** check box, the **Revoke All Licenses** option becomes available.
2. Once you have clicked on **Revoke All Licenses** the AVEVA License System Console performs the revocation of all licenses.

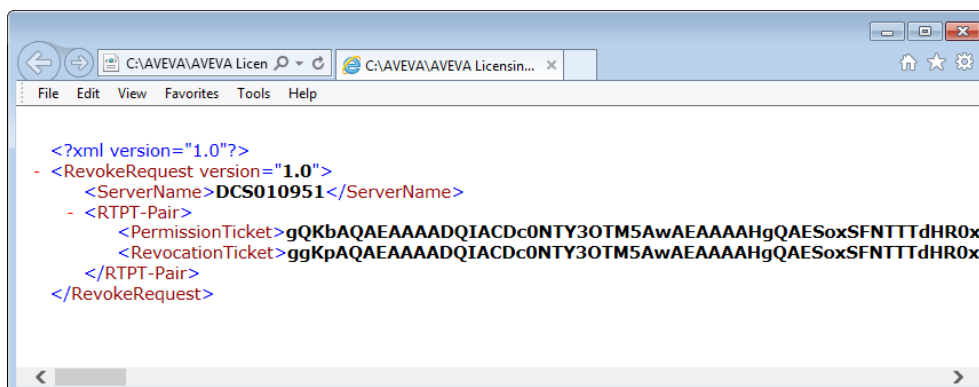
When the licenses are successfully revoked, the following message is displayed.



Disabled options can be re-enabled by clicking on any product or feature on the left side of the screen.

When the licenses are successfully revoked, the AVEVA License System Console saves the binary data of a Revocation Certificate and a Permission Ticket to an XML file.

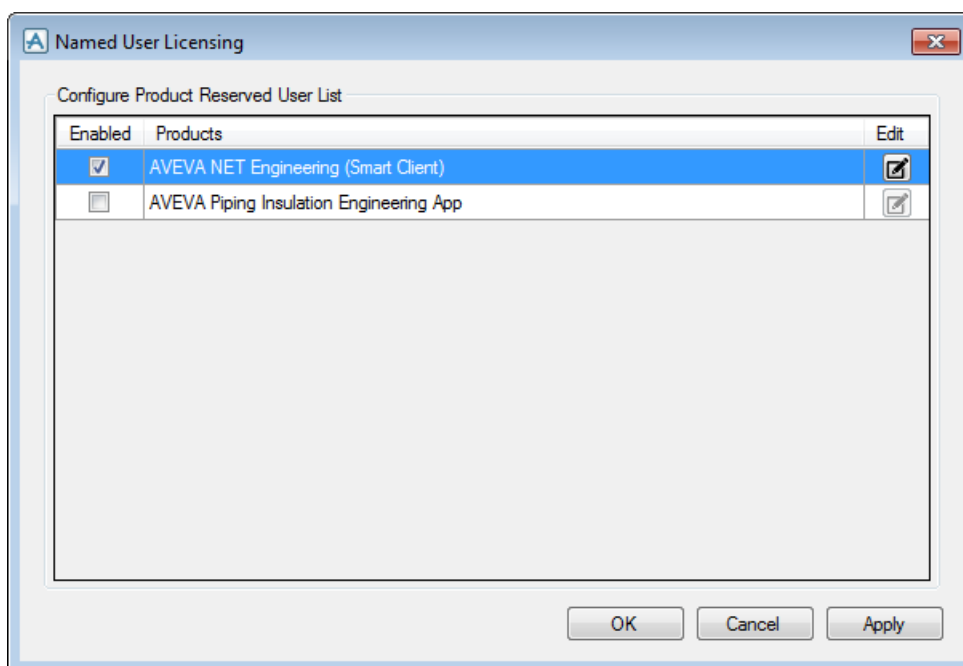
The AVEVA License System Console prompts you to send this file back to AVEVA as proof that the licenses were revoked.



Note: The number of Deferred Revocation Days is read from the Private Vendor Info of AVEVA "Meta Data License" which should be available on the license server. In cases where the value is not present or not valid (outside 1-30), a default value of 5 days is used.

Manage Named User License

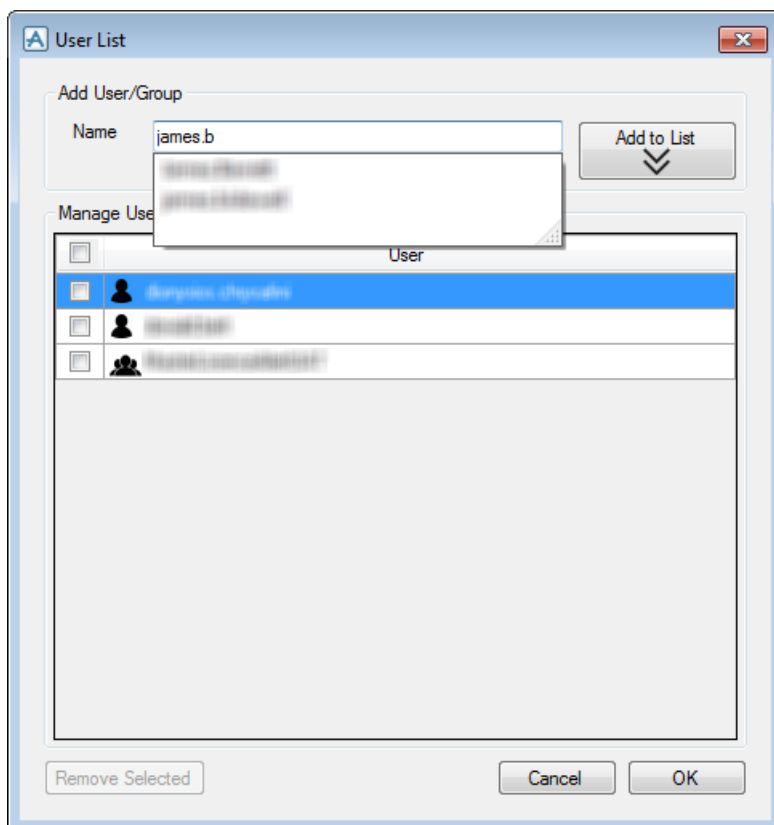
Clicking **Manage** opens a new window that allows you to reserve named user licences for a specific named user or group. A user list or group can be created for each available product.



1. Use the check boxes to select the products to associate with a named user list or group.  becomes available which allows you to modify the user list.
2. Click **Apply** to save the changes made to the window.

Modify User List

Clicking  opens a new window that allows you to add or remove named users or user groups from the reserved user list.

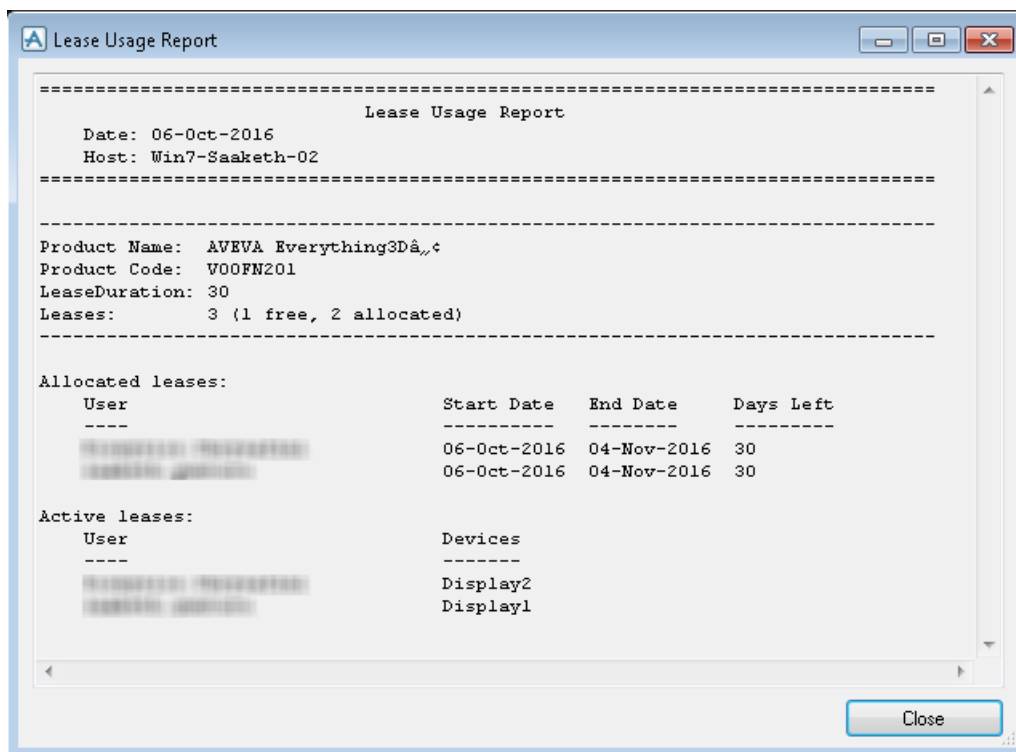


1. Input a valid user name or user group in the field. All matching user names and user groups become available from the drop-down list. The following rules should be observed when entering a user name or user group:
 - Ad-hoc windows user names are supported.
 - Active Directory (AD) users and groups are supported.
 - The LDAP (Lightweight Directory Access Protocol) sAMAccountName attribute is used to reserve licenses for user names, and the distinguishedName attribute is used for active directory groups.
 - Windows user aliases are not supported. Any aliases added to the reserved user list will block licenses from being used.
 - Using an AD group or an individual name (not both) is recommended. If a user is added explicitly and also belongs to an added AD group, the user count is doubled and two licenses are reserved.
 - AD groups that contain users that have left the company are automatically excluded.
 - Only windows username characters are supported, including "\" for specifying the domain name
2. Click **Add to List**: to add the user name or user group to the Manage Users list.

- When a specific user name or user group is selected via a check box, **Remove Selected** becomes available which allows you to remove the selection from the list.
- Click **OK** to apply the modifications to the user list.

View License Lease Report

Clicking **Lease Report** opens a new window that allows you to view the lease usage report for product licenses.



The lease usage report details the active leases and allocated leases for each licensed product.

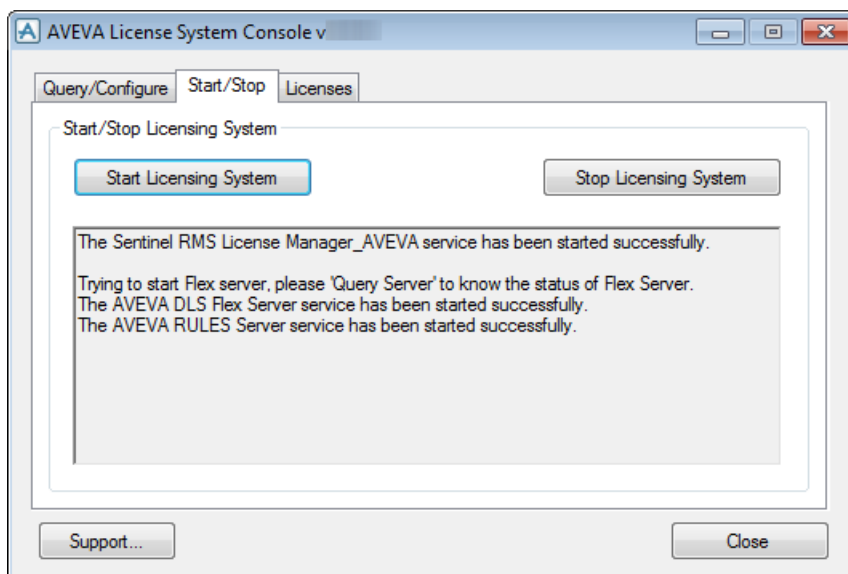
Support

Creates a directory that can be sent to AVEVA when a licensing support issue is raised. Refer to [Support](#) for further information.

2.2 Start/Stop

The **Start/Stop** tab allows you to start or stop the license servers. The messages area of the window displays the start/stop results so that any errors can be seen.

Note: It is not possible to stop or start just one license server.



Start Licensing System: Starts the RMS license server, the Flex license server and the RULES Server.

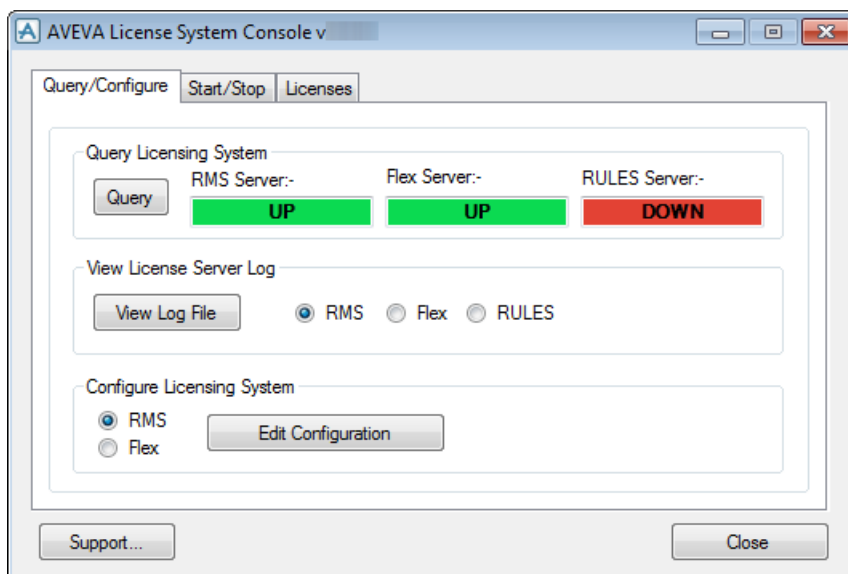
Stop Licensing System: Stops the RMS license server, the Flex license server and the RULES Server.

Support

Creates a directory that can be sent to AVEVA when a licensing support issue is raised. Refer to [Support](#) for further information.

2.3 Query/Configure

The **Query/Configure** tab displays the following window:



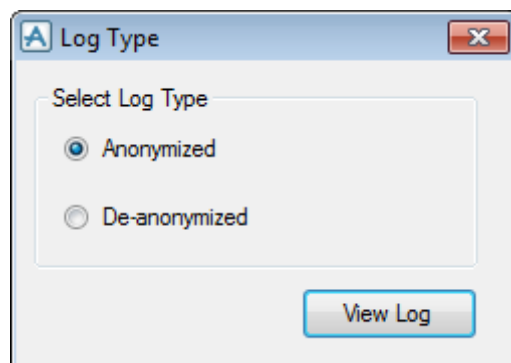
Query Licensing System

Upon clicking **Query** the status bars are updated to show the current status of each license server.

Note: The status boxes are not updated dynamically, update the status by clicking **Query** when appropriate.

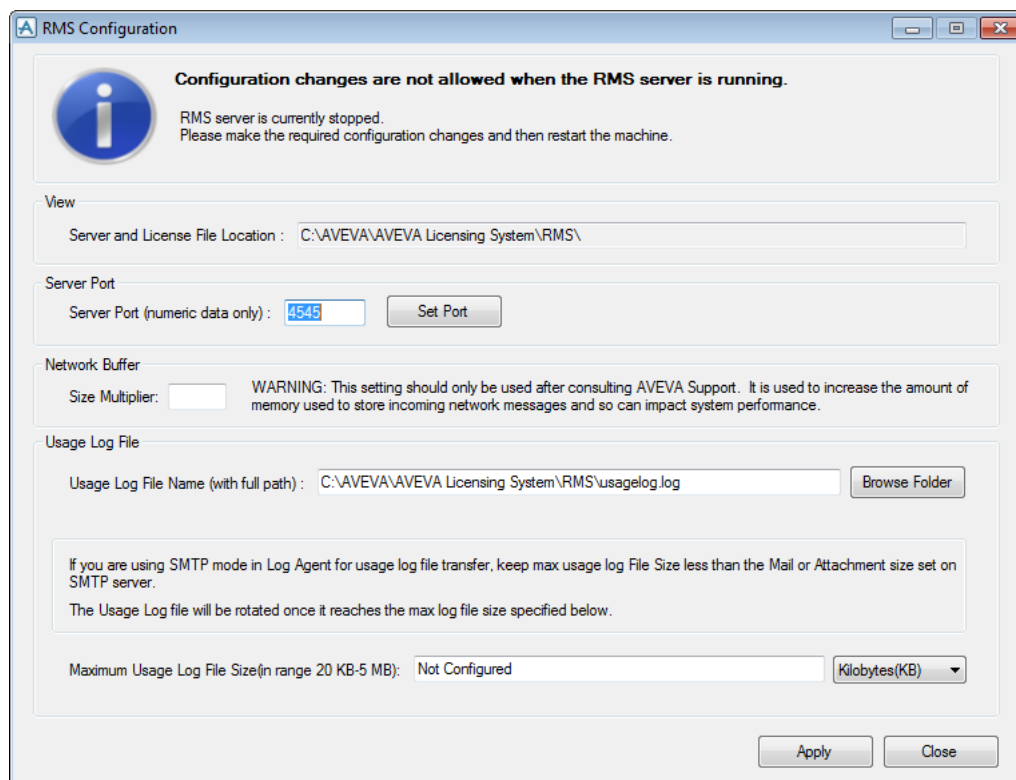
View Log

1. Use the radio buttons to select the required license server log file and click **View Log File**: a new window is displayed listing the contents of the license server log file.
2. If **RMS** is selected, an additional window is displayed that allows you to choose whether to view the default anonymised log file or the de-anonymised log file that displays the proper user, host, and display names.



Configure Licensing System

1. Use the radio button to select either RMS or Flex and click **Edit Configuration**.
2. If **RMS** is selected the following window is displayed:



- **Server and License File Location:** displays the path of the RMS license server and license file. This setting cannot be changed.
- **Server Port:** allows the RMS server port to edited.
- **Network Buffer:** The Network Buffer Size Multiplier is a network tuning parameter that enables the size of the RMS network buffer to be increased. The value is used as a multiplier for the existing Windows default buffer size which can vary from system to system (although for most systems it is usually 8 kBytes). Consequently, the actual buffer size used by the RMS server will not be known until the RMS server is started. Once started, the RMS server will write an event to the Windows Application Event log recording the actual size of the network buffer used.

Sites with a large number of users or with a heavily loaded network can experience instances of their AVEVA applications reporting "error 5". "Error 5" errors indicate that an application did not receive a response from a license server within the expected timeframe. This error can occur for several different reasons, however on sites with many users and/or with heavily loaded networks this problem can be mitigated by increasing the size of the socket buffer used by the RMS server.

There is no guidance available as to what value should be used for any given site as a suitable value must be determined by analysis of the site operating environment. It is therefore strongly recommended that this value is only configured after having consulted the AVEVA Licensing Department.

Important: Setting the network buffer value too high can impact performance, especially when using a license-server-list.

- **Usage Log File:** Allows the usage log file name and directory to be configured.

Note: The usage log file name cannot contain three or more continuous underscores.

Clicking **Apply** validates the entries made on the form and applies the updates to the configuration.

3. If the **Flex** radio button is selected, the Edit Flex Configuration window is displayed. The Flex Server Port can be changed. Clicking the **Edit** button opens the Flex configuration file (%AVEVA_LICENSING%\Flex\cadcentre.opt) in Notepad. You can then edit the file and save it.

Note:

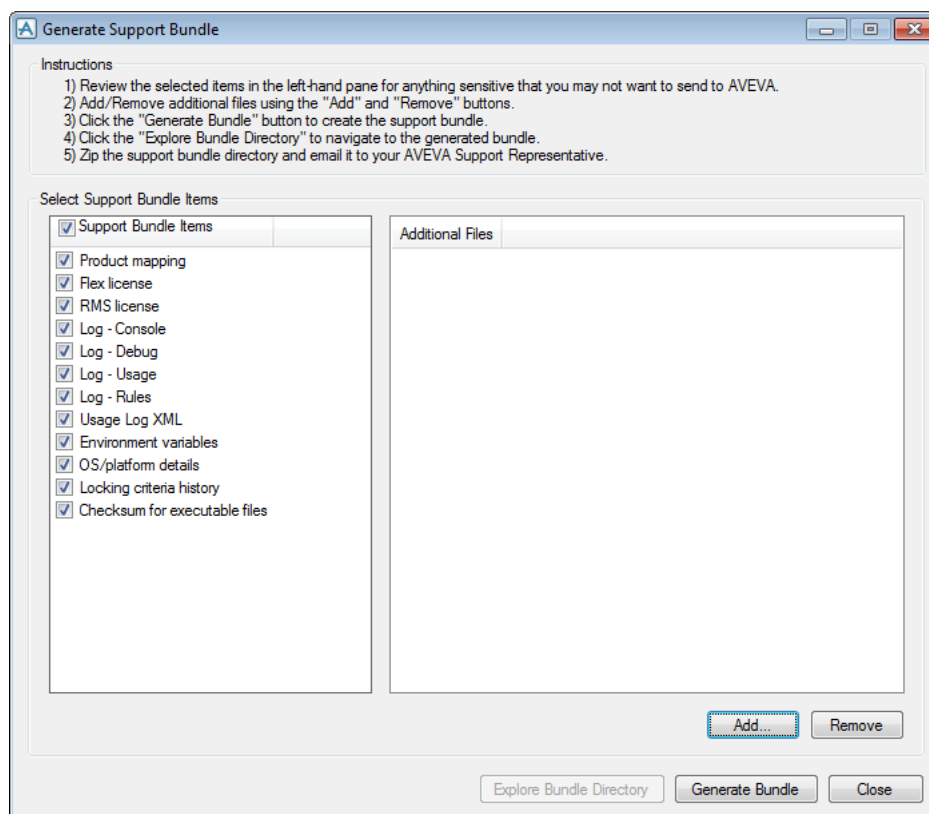
- No validation is carried out if you have changed the Flex configuration file.
- Changes made to the RMS log file configuration are only visible once the ALS console is restarted.

Support

Creates a directory that can be sent to AVEVA when a licensing support issue is raised. Refer to [Support](#) for further information.

2.4 Support

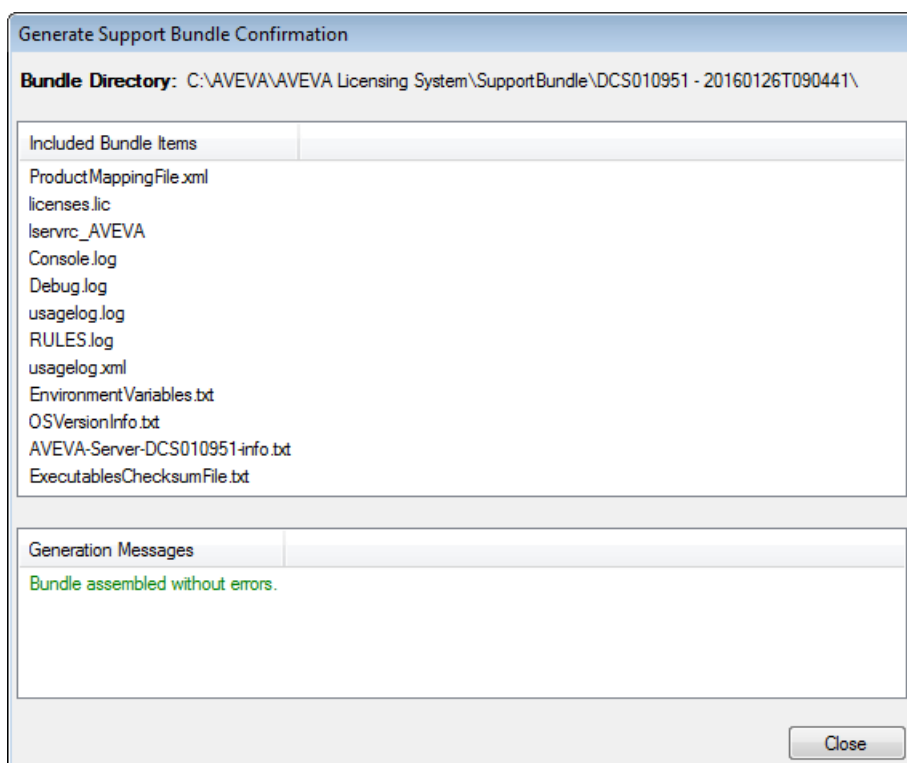
Clicking on **Support** from the main application opens a form that helps you create a support bundle consisting of a collection of licensing-related files from the license server that may be relevant to any issue or query you have. This facility creates a directory that can be easily zipped up and sent to AVEVA when a licensing support issue is raised.



The check boxes on the left-hand side of the window list a set of commonly collected files and system information that are likely to help with any support issue. By default, all items are selected. Any items that are not required can be excluded by deselection.

Additional files, such as screen shots or older logs, can be added by clicking **Add**. Each added file is listed on the right-hand side of the window.

Clicking **Generate Bundle** generates a dated folder located in %AVEVA_LICENSEING%\SupportBundle. All the selected items and additional files are copied into this folder, ready for zipping and sending to AVEVA support. The location and contents of the bundle is confirmed on the **Generate Support Bundle Confirmation** window; any issues in locating requested files are indicated with error messages.



You can navigate to the most recently generated folder by clicking **Explore Bundle Directory**.

A fresh folder is generated each time **Generate Bundle** is clicked. This is intended to make it easy to collect data at various points in a sequence of steps that are being followed to reproduce a problem. For example, if a problem occurs after starting an application and accessing a particular piece of functionality, the following workflow can be used to produce three separate support bundles, one before the application is started, one just before the problem occurs, and one after the problem has occurred:

1. Start the AVEVA License System Console and open the Support bundle form.
2. Select the files for inclusion in the bundle.
3. Click **Generate Bundle**. This creates the first bundle.
4. Start the application.
5. Click **Generate Bundle**. This creates the second bundle.
6. Access the suspect piece of functionality.

7. Click **Generate Bundle**. This creates the third bundle.

All three bundles can then be easily zipped and sent to AVEVA Support for analysis.

3 ALS Client Configuration Tool

License servers are identified by a hostname and a port number, this information must be configured so that your applications can function.

The ALS Client Configuration Tool is used to configure a workstation so that licensed applications can locate up to three AVEVA Licensing Systems and obtain the licenses required to function.

The Client Configuration Tool also allows configuration of a network timeout setting that can be used to tune performance on networks that offer intermittent connectivity to a license server. For example, a license server located on a remote site.

The Client Configuration Tool usually only needs to be used once per client workstation, when installing the first AVEVA application. However, it may be necessary to run the Client Configuration Tool again if the license server configuration changes.

Important: Before attempting to configure a workstation you must know the hostname and port number of each license server that is intended for use. If you do not have this information, contact your organisation's system administrator or IT department.

3.1 How License Server Details Are Configured

The license server configuration for a workstation is set using the environment variable AVEVA_LICENSE_SERVER_LIST. An environment variable permits you to configure applications to use different license servers by overriding the default workstation setting in a batch file.

License server host details are configured in the following format:

`<port>@<hostname>`

Where `<port>` is the port number used by the RMS license server and `<hostname>` is the hostname or IP address of the host on which the license server runs. A typical entry would look like this:

`4545@licserver`

Multiple AVEVA Licensing Systems can be configured by separating the details of each license server host with a semi-colon character (;):

`<port1>@<hostname1>;<port2>@<hostname2>;<port3>@<hostname3>`

A typical multiple server entry would look like this:

`4545@licserver1;4545@licserver2;4545@licserver3`

3.2 Client Use of Multiple License Servers

An AVEVA application always attempts to check-out a license from the first server in the AVEVA_LICENSE_SERVER_LIST. Only if that check-out fails, either because the server cannot be contacted or because it has no more licenses for a particular feature, does the application attempt to check-out from the next server in the list.

There are two reasons why you might configure a workstation with an AVEVA_LICENSE_SERVER_LIST pointing to more than one hostname:

1. Because you have three redundant license servers. Refer to [Three Redundant License Servers](#) for further information. If the first server fails, the client can connect to the second server and proceed using the same licenses.
2. Because, in consultation with AVEVA License Administration, you have deliberately divided your license entitlement across several, independent servers. For example, if you have users in several different locations, you can choose to install independent ALSs in several locations, with each client workstation attempting to connect to the local ALS first, then attempting to connect to the other ALSs if the local ALS has no licenses left.

3.3 Update a Workstation already Configured with AVEVA FlexMan Licensing

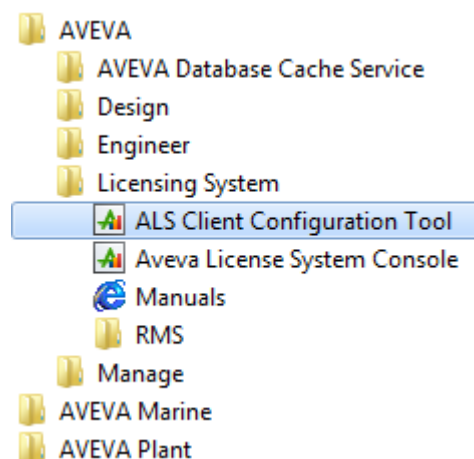
If a workstation is already configured to use FlexMan licensing, you will not need to reconfigure it if the new license server is installed on the same host as the original FlexMan license server and the new license server is configured to use the default port number of 4545.

Applications that support the AVEVA Licensing System are automatically detected and use FlexMan license server settings if the workstation has not been configured to use the AVEVA Licensing System.

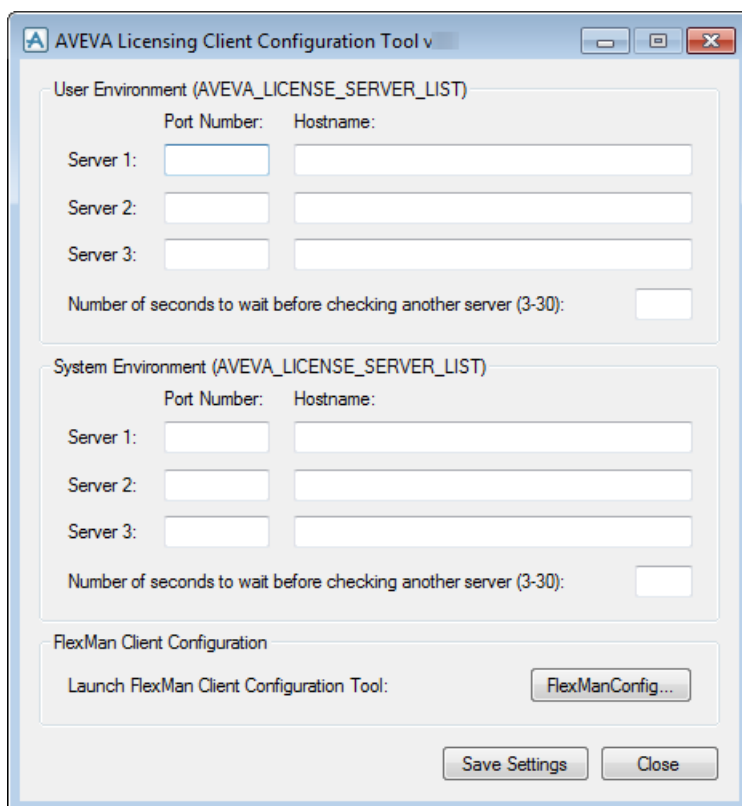
3.4 Configure a New Workstation

Configuration of a new workstation is carried out using the ALS Client Configuration Tool.

To launch the **ALS Client Configuration Tool**, select **Start > All Programs > AVEVA > Licensing System > ALS Client Configuration Tool**.



The **AVEVA Licensing Client Configuration Tool** window is displayed.



The tool allows the configuration of the license server and the timeout for the currently logged in Windows user (via the User Environment settings) or for all users on the workstation (via the System Environment settings).

FlexManConfig launches the FlexManConfig client configuration tool so that the workstation can be configured and applications licensed with FlexMan can be used.

After the ALS has been installed it is configured to use the following default ports:

RMS Server: Port 4545

Flex Server: Port 744

RULES Server: Port 4546

AVEVA recommends that the default ports are used if possible. If you do not want to use the default ports then they can be changed as follows:

RMS Server: a new port number can be entered using the **Edit Configuration** window on the **Query/Configure** tab of the **AVEVA License System Console**.

Flex Server: a new port can be specified by editing the last field on the **SERVER** line of the Flex license file. An editor (notepad) can be started by clicking **View License File** on the **Licenses** tab of the **AVEVA License System Console**).

RULES Server: Always uses the port number after that of the RMS Server.

If AVEVA applications have previously been installed, it is highly recommended that you use the same port numbers configured in the workstations that run the AVEVA applications. To find out these port numbers, implement the following:

- For workstations running AVEVA applications that are licensed with the ALS, run the ALSClientConfigTool.exe tool.
- For workstations running older applications that are licensed with the FlexMan licensing system, run the FlexManConfig.exe tool.

When the license server port numbers have been set, the workstations that run the applications must be configured so that they can locate the license servers. To do this, run either ALSClientConfigTool.exe or FlexManConfig.exe depending on whether the applications are licensed with ALS or FlexMan.

Important: If the workstations are already configured with settings for a FlexMan license server, you can avoid the need to reconfigure the workstations by:

1. Installing the ALS on the same host as the old FlexMan server (which must be uninstalled first).
 2. Using the default RMS port number 4545.
-

New applications that are licensed with ALS RMS will automatically use the older FlexMan settings if no ALS settings are configured. In these circumstances, applications attempt to communicate with an ALS RMS server on port 4545 on the same host as the configured FlexMan server host. For example, if there is an existing FlexMan server on a host called **myhost** and the workstations are configured with the FlexMan setting **744@myhost** then you can install the ALS server on **myhost**, set the RMS server port to 4545 and install the ALS enabled applications on the workstations without the need to reconfigure them.

4 Sentinel RMS License Manager Administration

The Sentinel RMS License Manager Administration tool allows you to define group reservations which are used to restrict the right to run applications for particular users, computers, or groups. Selected users can be allowed to run certain applications or can be prevented from accessing applications. This can be used as an additional security measure and helps to balance application usage when the administrator wants to prevent users from running applications on heavily loaded computers or file servers.

This allows the administrator to exercise local administrative controls by associating a series of groups with each feature and reserving a certain number of license tokens for each group. Any remaining license tokens are not specifically reserved and can be used by anyone.

Note: The group reservations capability is only used with a network License Manager. It is not supported when the redundant License Manager pool is set up.

4.1 Reservation File Characteristics

The reservation file indicates the amount of license tokens that are reserved for members of different groups. For example, the Sales department could be a group of ten members for which 15 license tokens are reserved. The members might be network users or specific computers on the network. The reservation file might also indicate that certain members of the group are excluded from taking license tokens.

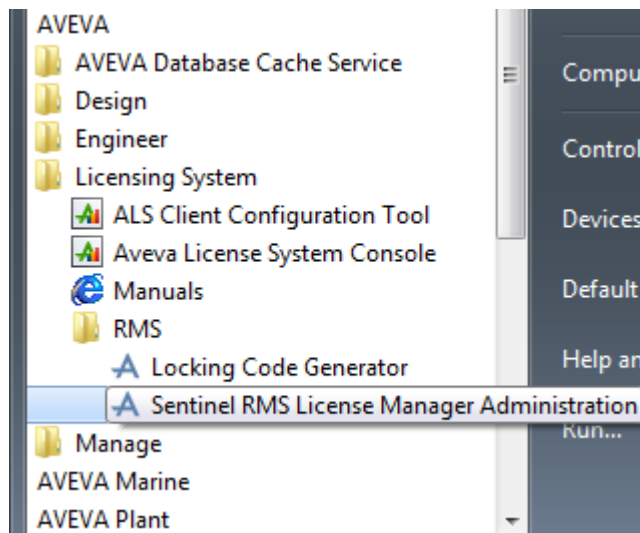
When the License Manager receives a request, it checks whether the user making the request belongs to a group. If so, and license tokens are available for that group, the License Manager issues the token(s) and decrements the token count on the License Manager. Otherwise, requests will be serviced with tokens that have not been reserved and that are available.

The following information applies:

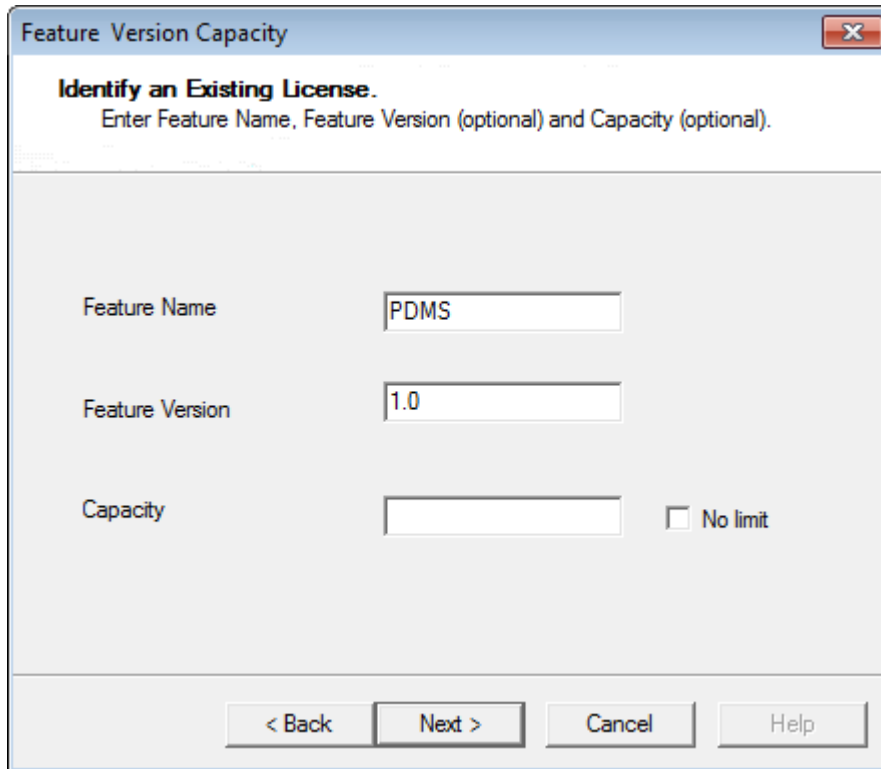
- A maximum of 256 groups, each with 1000 members are supported.
- Groups must be mutually exclusive. Different groups for the same feature must not have common users or computers.
- Group names and member names cannot exceed 64 characters each.
- The number of license tokens reserved for a feature cannot exceed the number of concurrent copies specified in the license code for that feature. If the number of license tokens reserved is greater than the number of concurrent copies specified for the feature in the license code, the number of tokens specified in the license code is used.
- To exclude all members of a group from using license tokens for a feature, specify 0 tokens.

4.2 Creating the Reservation File

1. To launch the **Sentinel RMS License Manager Administration** tool, select **Start > All Programs > AVEVA > Licensing System > RMS > Sentinel RMS License Manager Administration**.



2. Select **Reservation File** from the **Edit** menu.
3. To make a new reservation file, select **New** from the **File** menu or select **Open** to edit an existing reservation file.
4. To add user definitions for license tokens, select **Add** from the **Feature** menu. The **Add License Reservation Wizard** is displayed.
5. Click **Next**. Enter the feature name=XXX, feature version =1.0 and click **Next**.



Feature Version Capacity

Identify an Existing License.
Enter Feature Name, Feature Version (optional) and Capacity (optional).

Feature Name:

Feature Version:

Capacity: ☐ No limit

< Back Next > Cancel Help

Note: Do not populate the **Capacity** field.

- Enter a Group name and the number of reserved Tokens for that group then click **Next**.

Group

Add a group for the specified non-capacity feature.
Enter the group name and number of tokens to be reserved for the group

Group Name:

Tokens:

< Back Next > Cancel Help

7. After specifying the above details, the Wizard prompts you to specify the members of the group. Click **Add** to add a member.
8. Enter the Name of the Member, then select **User** or **Machine** and select **Included** or **Excluded** to specify whether member is allowed or not allowed to run the application.

Members

Add members to the group.
To add a member

Member Name

Add...

Member

Name of the Member:

Members of a group can either be Users or Machines. This member is a :

☒ User
☐ Machine

Members can be allowed to run the application by Including them Or can be prevented to run the application by excluding them. This member is :

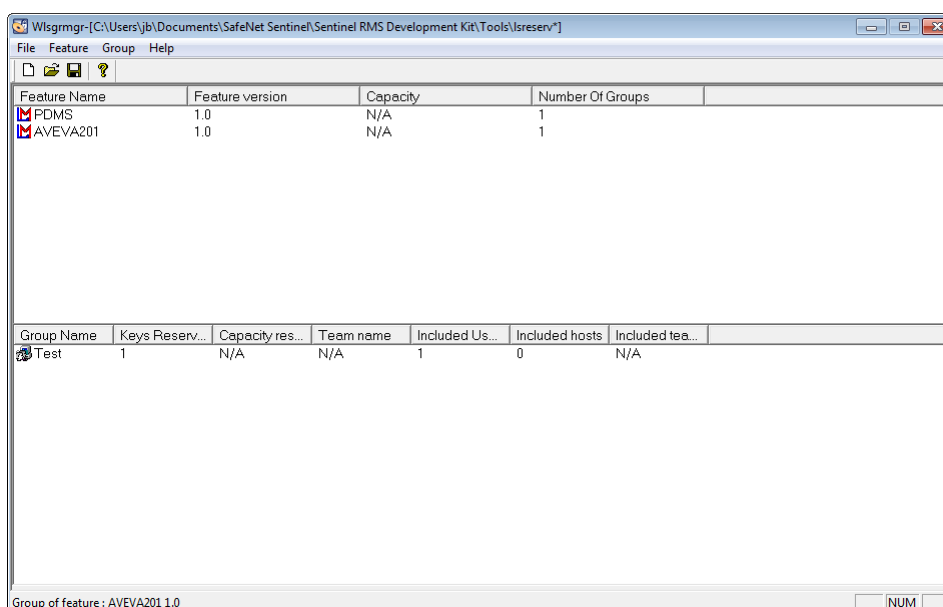
☒ Included
☐ Excluded

OK Cancel

< Back Finish Cancel Help

9. Click **OK** to finish the definition.

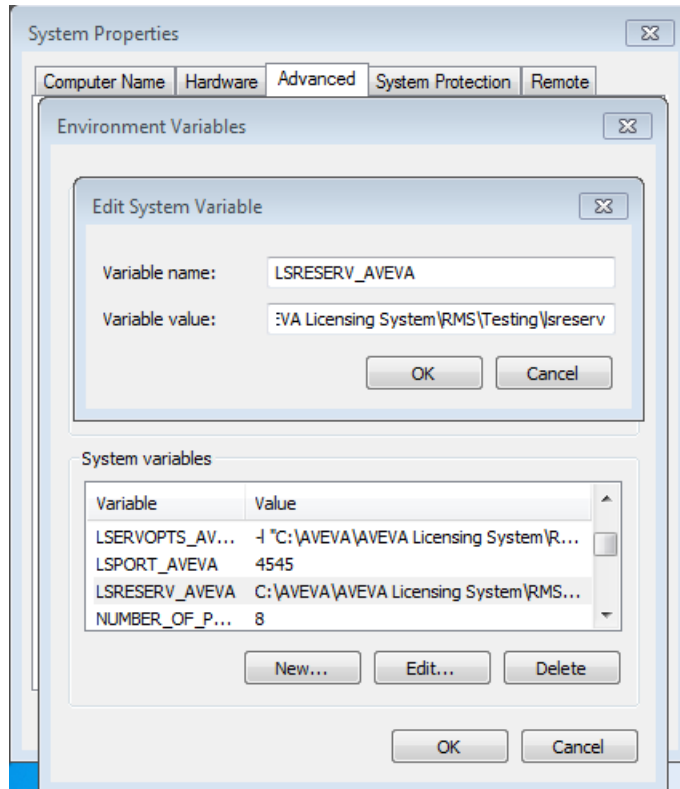
10. If required, click **Remove All** to remove all the members or select a specific member and then click **Remove**.
11. Click **Finish** to return to the main screen, the groups and members are updated on the WLSGRMGR main screen.
12. Once a feature has been defined, you can define another group to use that feature. To add a new group select **Add** from the **Group** menu. Do not add the same users or computers to different groups using the same feature.
13. Click **Save** to save the settings: a reservation file 'lsreserv' is created in the My documents folder 'C:\Users\Administrator\Documents\SafeNet Sentinel\Sentinel RMS Development Kit\Tools'.
14. Copy the 'lsreserv' file into the RMS folder 'C:\AVEVA\AVEVA Licensing System\RMS' and rename it 'lsreserv_AVEVA'.



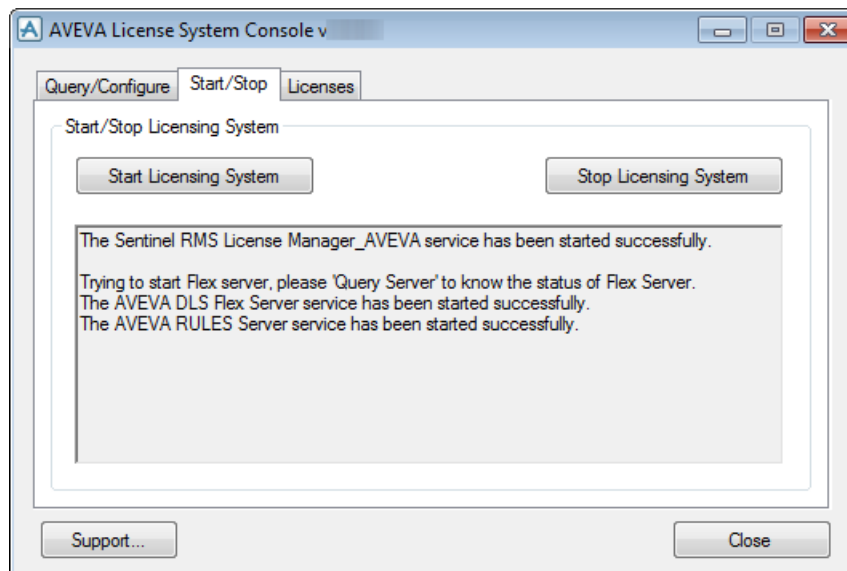
15. To save the file directly into the RMS folder 'C:\AVEVA\AVEVA Licensing System\RMS', click **Save As**.

Or

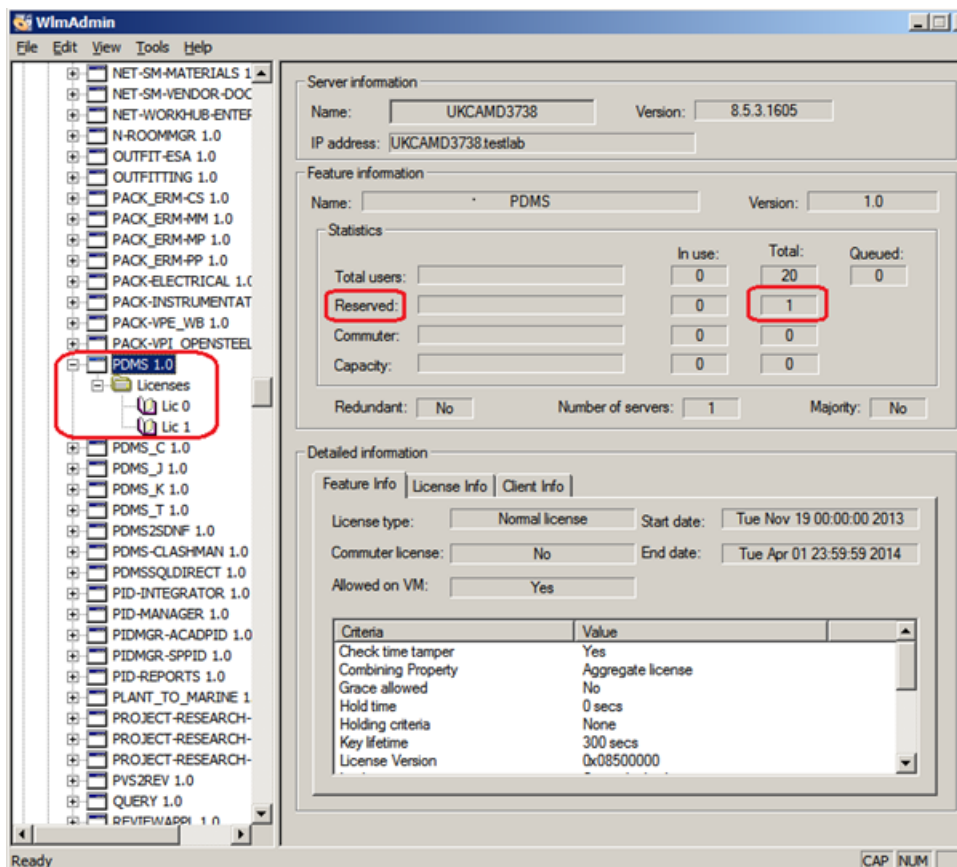
If you copy the reservation 'lsreserv' file to a folder other than the RMS folder, an environment variable must be set, that is Variable Name = LSRESERV_AVEVA and Value = The folder path to the lsreserv file. For example 'C:\AVEVA\AVEVA Licensing System\RMS\Testing\lsreserv'.



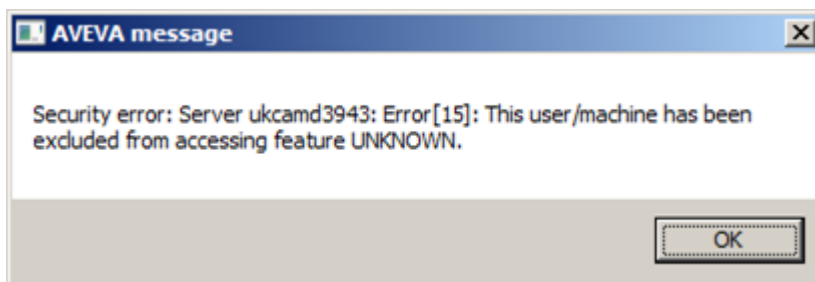
16. Once the environment variable has been set, restart the License Server using the **AVEVA License System Console**.



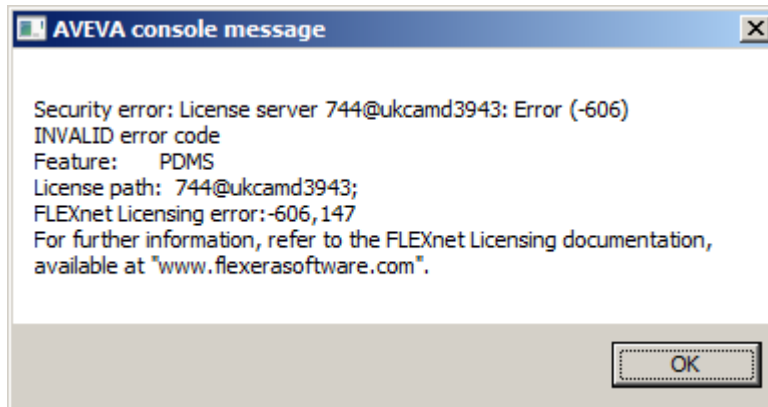
17. The number of licenses reserved for the particular feature can be viewed using the Sentinel RMS License Manager Administration tool.



18. If you run a RMS secured program, for example AVEVA E3D, which has been reserved and its usage restricted, the following error message is displayed.



19. If you run a Flex secured program, for example Hull Design or PDMS 12.0, which has been reserved and its usage restricted, the following error message is displayed.



Sample Reservation file contents:

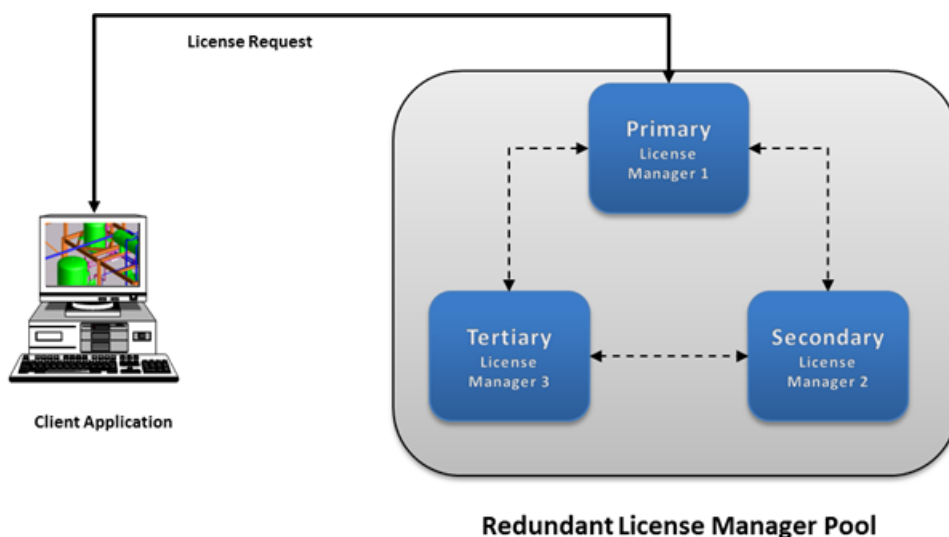
AVEVA201,1.0:Test:1:User1 !User2

This means:

- Feature Name - Aveva201
- Feature Version - 1.0
- Group Name - Test
- No of Seats Reserved - 1
- License Reserved for the user - User1
- License disallowed for the user - User2

5 Three Redundant License Servers

To work with the redundant license model, a License Server pool must be set up. The primary License Server in the pool is designated as the leader; the remaining License Servers become followers. As long as the primary License Server is running, all the tokens remain with the leader License Server and are not distributed amongst the other License Servers in the pool.



How it works

- A client sends a license request to the primary License Server.
- The primary License Server establishes a connection with the client and grants the license after successful authentication.
- When the primary License Server (leader) is not running, the next leader takes over: the follower License Server having the highest precedence.
- The follower License Server fulfils the new license requests seamlessly (as long as licenses are available).
- If the primary License Server re-starts, it reclaims leadership and license control is transferred back to the primary License Server.

5.1 Setting up Redundant License Servers

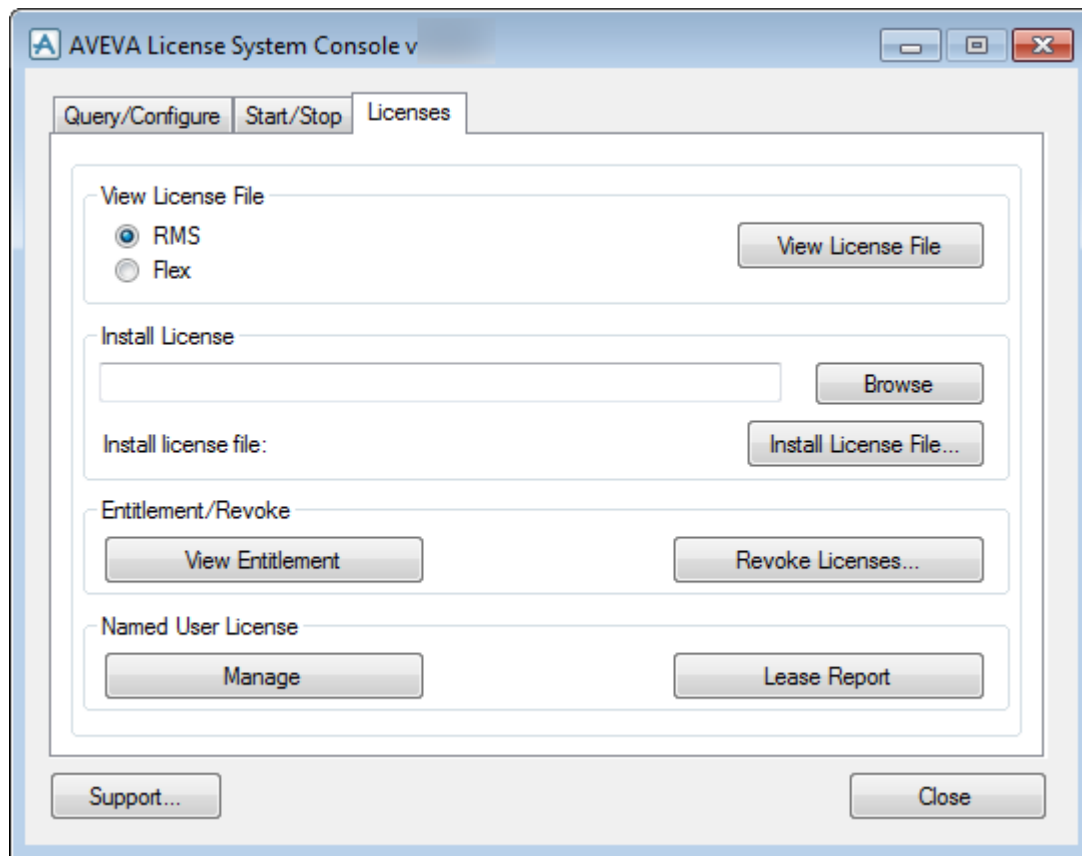
The following requirements and recommendations are for system administrators:

1. First, select the computers on which the License Servers will reside.

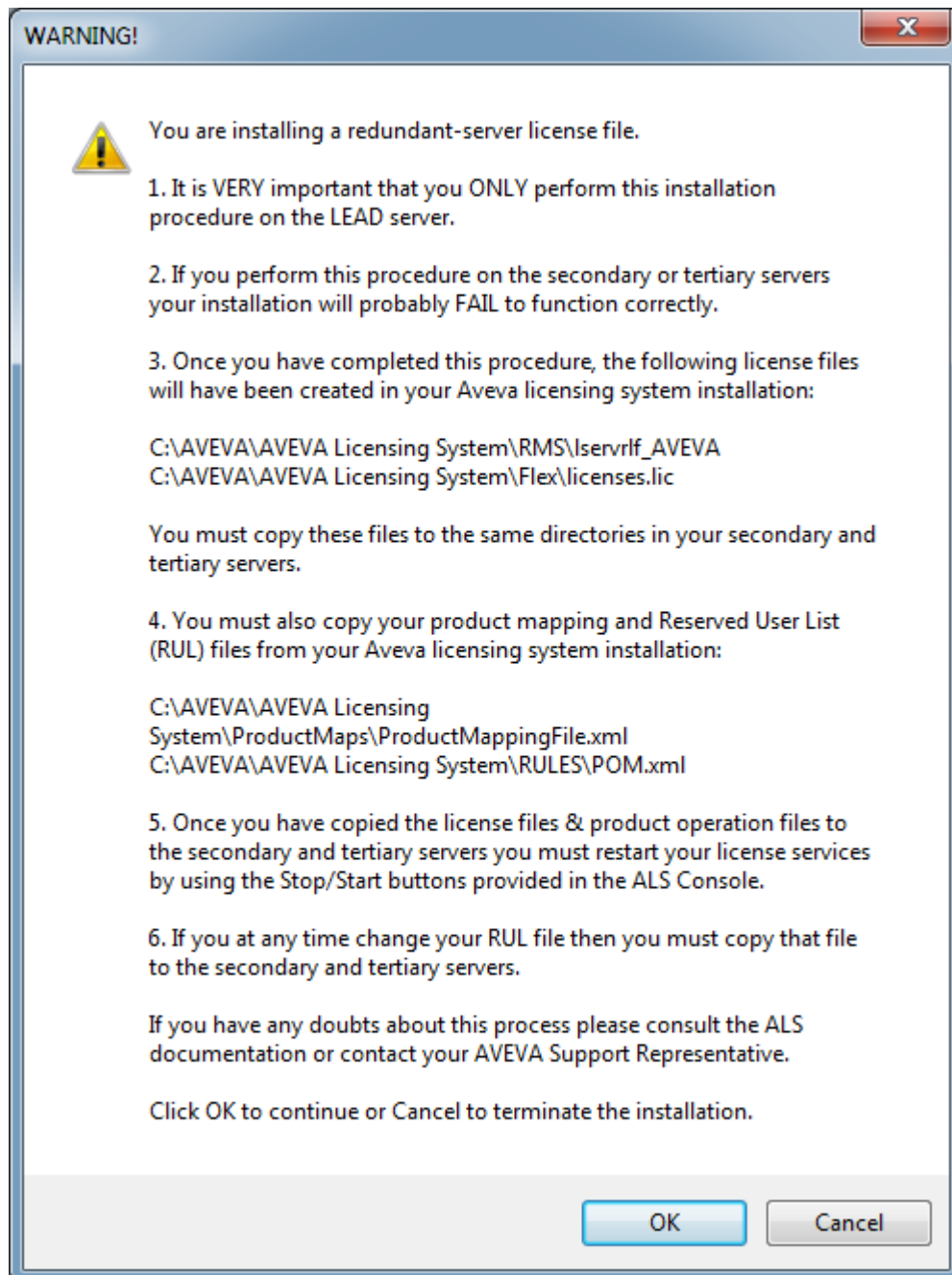
2. Various factors, including network performance, affect this decision. The leader should be in an area of the network with good bandwidth because it must communicate with all other redundant License Servers,
3. The system administrator must define at least three redundant License Servers. If the pool is set up with only a single server, the redundant server pool will never be formed.
4. A License Server can only exist in one license pool.
5. Each computer on which a redundant License Server resides must have a static, fixed IP address. Do not use DHCP to dynamically allocate IP addresses for those computers.
6. Reverse DNS resolution is necessary for the redundant License Server setup.
7. Specify the IP addresses in either IPv4 or IPv6 format. A redundant pool cannot have mixed servers (both IPv4 and IPv6). Consequently, a redundant server can work either in IPv6 or IPv4 mode-as long as the server is running and not removed from the redundant pool.
8. After establishing the License Servers for redundant setup, obtain their locking codes and pass them to the AVEVA license administrator.
9. Setting up the redundant License Servers requires access privileges to all computers and areas of the network on which License Servers are to be installed. If changes are made to the redundant license file, those changes will not be transferred to all License Servers in the redundant License Server pool unless network and write access to each of the redundant License Server computers is allowed. If any of these computers run Windows, the system administrator must have administrator privileges to make changes that affect the License Servers on those computers.
10. The License Servers in the redundant License Server pool must have the same version.
11. The License Servers must preferably be running on the same platforms (for example, Windows).

5.2 Configuration and installation of the Three Redundant Server License Files

1. Install the ALS on all License Server pool machines.
2. Generate a locking information file on all machines by double clicking on the AVEVA-locking.bat file. The bat file can be found in the Aveva\Aveva Licensing System\RMS\Tools\ folder.
3. Send all locking information files to the AVEVA license administrator and specify which machines are to be the primary (leader), secondary, and tertiary license servers in the License Server pool.
4. Save the license file, received from the license administrator, on the primary (leader) server machine.
5. Run the AVEVA License System Console to install the license file on the primary (leader) server. To launch the **AVEVA License System Console** select **Start > All Programs > AVEVA > Licensing System > AVEVA License System Console**.



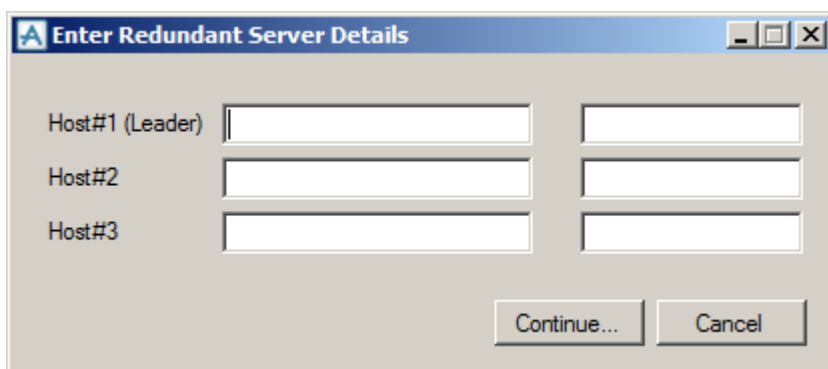
6. Click the **Licenses** tab.
7. Click **Browse** and select the license file.
8. Click **Install License File**, the following message is displayed.



Steps 4 & 5 of the warning suggest that you copy the Restricted User List file (POM.xml) to the other License Servers. This file defines how leases are allocated for named user licenses.

- An identical Restricted User List file must be used on each License Server in the License Server pool.
- Subsequent changes made to the Restricted User List file, on any of the License Servers, must be manually copied to all License Servers in the License Server pool.

- The Restricted User List file of all License Servers in the License Server pool must not be different at any time. This may result in the three redundant server setup behaving abnormally.
9. Click **Continue**
 10. Enter system server host names and IP addresses (the IP addresses are populated automatically but should be double-checked as some hosts can have multiple IP addresses).

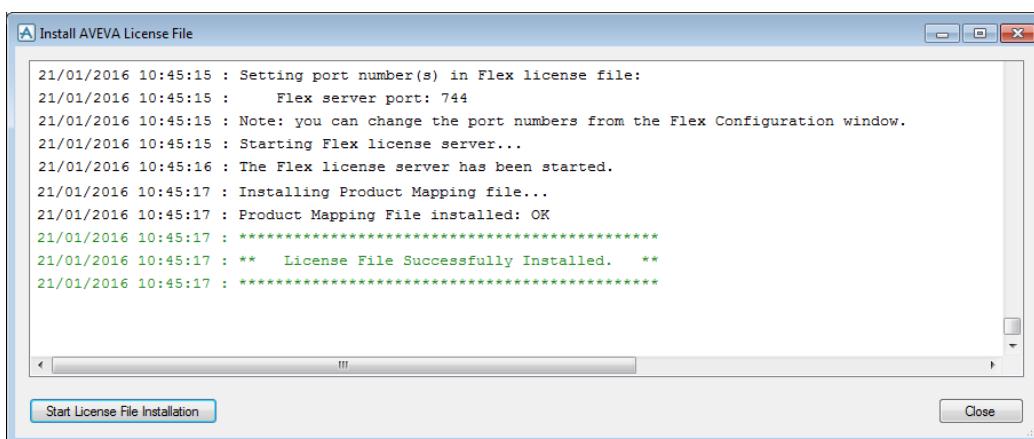


The dialog box titled "Enter Redundant Server Details" contains three rows of input fields for Host#1 (Leader), Host#2, and Host#3. Each row has two adjacent text boxes for host name and IP address. At the bottom right are "Continue..." and "Cancel" buttons.

Note: Server details should be entered in the same order as you request the license file (refer to 3. for further information).

11. Click **Continue**.
12. On the **Install AVEVA License File** window, click **Start License File Installation**.
13. Click **Yes** in the confirmation window.

The license file installation success/failure status is displayed in the **Install AVEVA license file** window.



The "Install AVEVA License File" window shows a log of installation steps. The log text is as follows:

```

21/01/2016 10:45:15 : Setting port number(s) in Flex license file:
21/01/2016 10:45:15 :     Flex server port: 744
21/01/2016 10:45:15 : Note: you can change the port numbers from the Flex Configuration window.
21/01/2016 10:45:15 : Starting Flex license server...
21/01/2016 10:45:16 : The Flex license server has been started.
21/01/2016 10:45:17 : Installing Product Mapping file...
21/01/2016 10:45:17 : Product Mapping File installed: OK
21/01/2016 10:45:17 : *****
21/01/2016 10:45:17 : **  License File Successfully Installed.  **
21/01/2016 10:45:17 : *****
  
```

At the bottom left is a "Start License File Installation" button, and at the bottom right is a "Close" button.

Note: The checks and installation details of the license file are written to a Console.log file and stored in the default folder (C:\AVEVA\AVEVA Licensing System). If there is a failure in the license file installation, send the Console.log file to the AVEVA license administrator or the AVEVA support engineer.

14. Once you have successfully completed the license installation procedure on the primary server, the following files are created in your AVEVA licensing installation (by default C:\AVEVA\AVEVA Licensing System).

C:\AVEVA\AVEVA Licensing System\RMS\servrlf_AVEVA.

C:\AVEVA\AVEVA Licensing System\Flex\licenses.lic.

15. Copy these files to the same directories on your secondary and tertiary servers.
16. Once you have copied the license files, restart your license services by using the Stop/Start buttons provided in the ALS Console.

Important: You must ONLY perform the license installation procedure (from [5.](#) to [13.](#)) on the PRIMARY server, and copy the generated files onto the secondary and tertiary servers as described. If you perform the license installation procedure on the secondary or tertiary servers, the licensing system will fail to function correctly.

6 RULES SERVER

The RULES server provides support for an AVEVA licensing model called "Named User Licensing". Named User Licensing differs from the floating license model in three major aspects:

- Licenses are restricted to a specific number of named users.
- Licenses are locked to a specific user for a fixed number of days from "first use". For example, 30 days.
- One Locked license can be used by a user on a limited number of devices.

Named users are identified automatically by the RULES server as the first users to request a license. Once all the licenses are in use, no further users can connect until the end of the 30 day period for one of the licensed users.

In effect, this is a form of reserved user licensing, hence the acronym: RULES - Reserved User Licensing Enforcement System.

6.1 RULES System and Port Configuration

The operation of RULES is fully automatic and no configuration is required. RULES obtains its operating parameters directly from the RMS license system, there are no additional duties for an administrator other than opening a port for the RULES server on the system firewall.

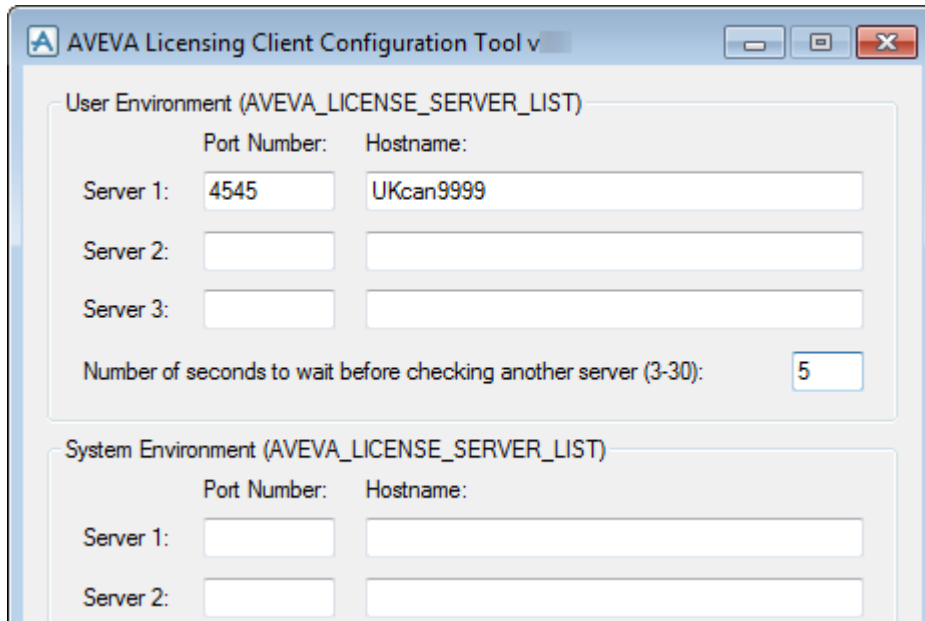
The RULES server uses its own port to communicate with RULES applications (for example: AVEVA Insight). The port number is derived from the RMS server port number thus: the RULES server uses the next highest port after the RMS server port. For example, if the RMS server is configured to use port 4545, then the RULES server will automatically use port 4546. The RULES port number cannot be set independently of the RMS port number.

6.2 RULES Client Configuration

RULES clients automatically determine the RULES server port number by using the same calculation as the RULES server, using the configured RMS server port number plus 1.

For example, if a client is configured with a server setting of 123@myServer, then the RULES client attempts to communicate with a RULES server on port 124 on the server called myServer. If the ALS client is already configured to the RMS server port then there is no need to change the client settings.

Example: In the ALS client configuration tool, the administrator must mention the RMS server port (by default 4545) and license server name, that is 4545@servername. The RULES client then automatically communicates with the RULES server on port 4546.



AVEVA Licensing Client Configuration Tool v

User Environment (AVEVA_LICENSE_SERVER_LIST)

	Port Number:	Hostname:
Server 1:	4545	UKcan9999
Server 2:		
Server 3:		

Number of seconds to wait before checking another server (3-30): 5

System Environment (AVEVA_LICENSE_SERVER_LIST)

	Port Number:	Hostname:
Server 1:		
Server 2:		

6.3 RULES as a proxy to RMS

For ALS version 2.0, RULES was an adjunct license server used only for Named User licenses.

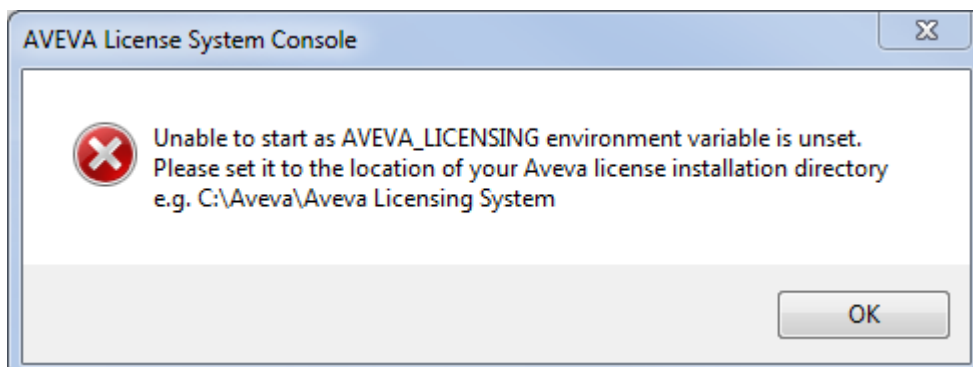
From ALS version 3.0 and RULES enabled applications, the RULES license server acts as a proxy to the RMS server for all licenses. This applies to all licenses, not just Named User licenses. One advantage of this approach is that the connection between the application and the ALS is TCP with RULES, rather than UDP directly with the RMS license server.

7 Frequently Asked Questions

To help troubleshoot any problems AVEVA have compiled a list of frequently asked questions.

7.1 What should I do if the error "Unable to start as AVEVA_LICENSING environment variable is unset" is displayed?

When running the AVEVA License System Console for the first time, the following error - Unable to start as AVEVA_LICENSING environment variable is unset is displayed:

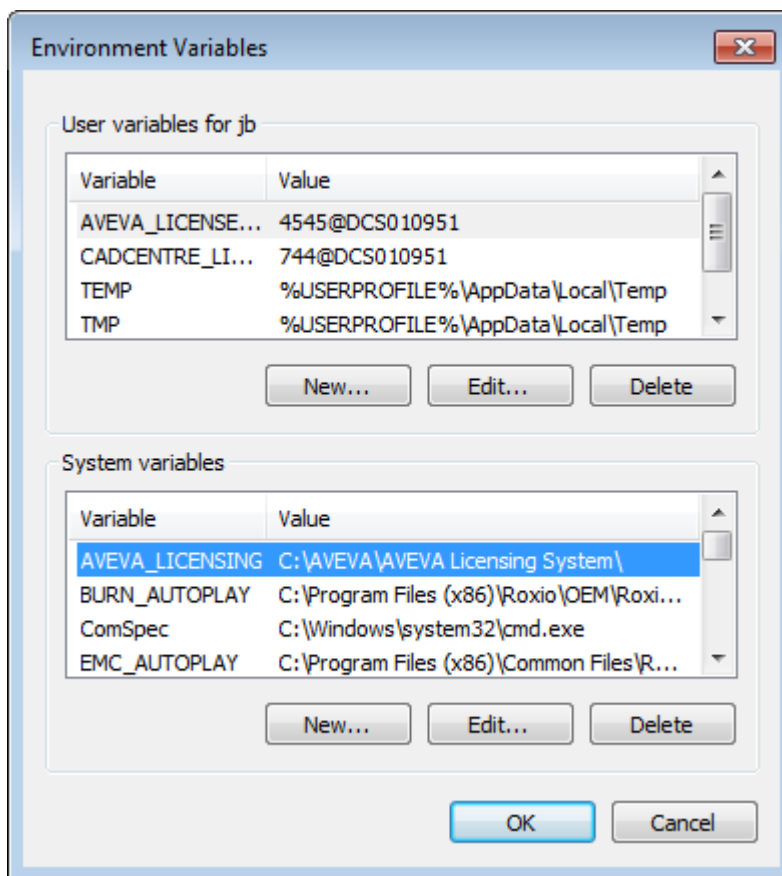


This can happen on Windows servers. It is a Windows timing problem, the installer has set an environment variable but its value is not populated. As a result, various workarounds are available.

Wait for a few minutes and try again

or (reboot) the machine

Or



Navigate to the environment variable, select 'AVEVA_LICENSING' > Click **Edit** then click **OK** to set it again.

7.2 How to generate the ALS Server Locking Criteria?

Run the **AVEVA-Locking.bat** file, this can be found in the path 'C:\AVEVA\AVEVA Licensing System\RMS\Tools'. This creates a file called **AVEVA-Server-HOSTNAME-info.txt**.

Email this file to the AVEVA License Administrator along with the server product requirements.

The license administrator will return a license file for that server.

7.3 What should I do if AVEVA-Server-name-info.txt file does NOT have a UUID?

When running the **AVEVA-locking.bat** file the **AVEVA-Server-name-info.txt** file does NOT have a UUID listed. For example:

Sentinel RMS Development Kit 8.5.3.0020 Host Locking Code Information Utility Copyright © 2012 SafeNet, Inc.

```

IP address           : 192.168.200.56
Disk ID             : 0xCD341ADF
Host name           : UKCAMD3903
Ethernet address[1] : 18-A9-05-C2-37-21
Hard Disk Serial[1] : 9VMC5AEK
CPU Info String     : GenuineIntel Intel(R) Core(TM) i7 CPU Q 740 @
                    : 1.73GHz 6 14 5
UUID                : (This is returned Blank No UUID listed)
  
```

This sometimes happens on Windows 2003 servers if the ALS system has been installed and the system has NOT been re-started.

Once the server is restarted, run the **AVEVA-locking.bat** file, a UUID value is returned. A UUID value is required in order to obtain a license from AVEVA that will work.

7.4 What should I do if AVEVA-Server-name-info.txt file does NOT have an Ethernet address?

When running the **AVEVA-locking.bat** file the **AVEVA-Server-name-info.txt** file does NOT have an Ethernet address listed. For example:

Sentinel RMS Development Kit 8.5.3.0020 Host Locking Code Information Utility Copyright © 2012 SafeNet, Inc.

```

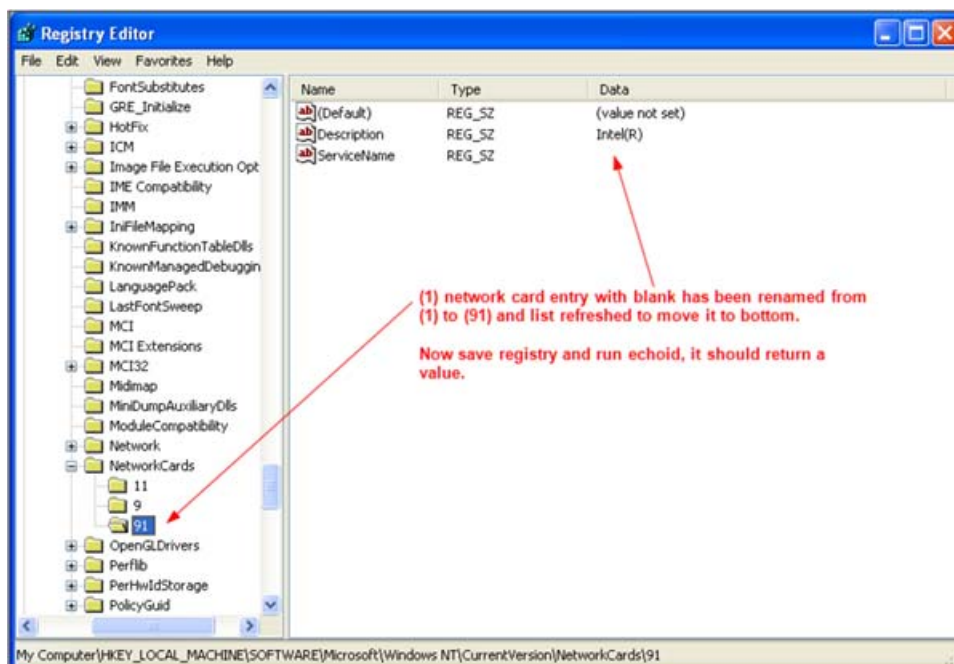
IP address           : 192.168.200.56
Disk ID             : 0xCD341ADF
Host name           : UKCAMD3903
Ethernet address[1] : (This is returned Blank No Ethernet address)
Hard Disk Serial[1] : 9VMC5AEK
CPU Info String     : GenuineIntel Intel(R) Core(TM) i7 CPU Q 740 @
                    : 1.73GHz 6 14 5
UUID                : BD5E286E-30D0-11DF-BBDA-05C2372118A9
  
```

In the registry, the first network card listed has a blank in it, or is a network card that the RMS does not use. For example, a Phone card/mini network card. When this happens, instead of looking for the next entry echoid/wechoid stops and returns a blank. To fix this problem, you must carefully edit the registry and move the first network card entry to the bottom of the list, thus enabling the echoid/wechoid programs to pick up a valid Ethernet address and list it.

Open the machine registry and navigate to:

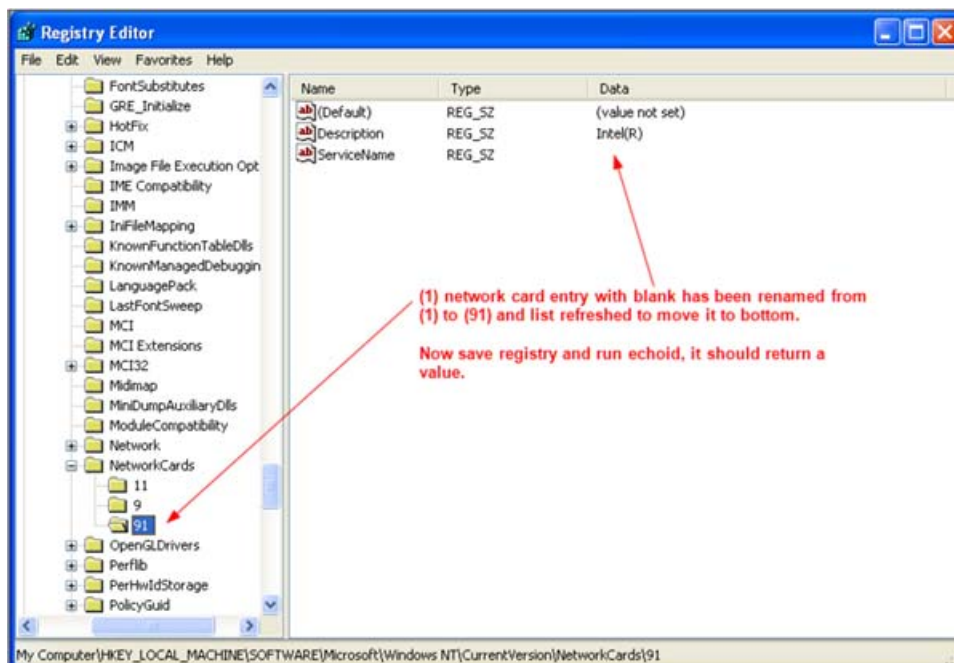
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\WindowsNT\CurrentVersion\NetworkCards

The following example shows a blank entry:-



To move the blank or problem network card to the bottom of the list, rename the number to a higher network card number than the last one in the list.

In this example, it is renamed to 91 but it could have been 12.



Once the entry has been changed, the list refreshed, the registry closed, and the **AVEVA-locking.bat** file run, the **AVEVA-Server-name-info.txt** file output contains an Ethernet address. For example:

Sentinel RMS Development Kit 8.5.3.0020 Host Locking Code Information Utility

Copyright © 2012 SafeNet, Inc.

```
IP address           : 192.168.200.56
Disk ID             : 0xCD341ADF
Host name           : UKCAMD3903
Ethernet address[1] : 18-A9-05-C2-37-21
Hard Disk Serial[1] : 9VMC5AEK
CPU Info String     : GenuineIntel Intel(R) Core(TM) i7 CPU Q 740 @
                    : 1.73GHz 6 14 5
UUID                : BD5E286E-30D0-11DF-BBDA-05C2372118A9
```

Now a license file can be requested using the Ethernet address.

Note: If the license file is requested with a network card active, when the returned license.xml file is installed, it MUST be installed with the same network card active.

7.5 What should I do if I get the error 'locking-code mismatch' during the license file installation?

This means that the locking criteria, supplied to AVEVA to generate the license, does not match the current hardware characteristics of the license server. A customer's license can be locked to many criteria, and only the license administrator can inform you of the criteria.

7.6 Where is the composite locking criteria code in ALS license file?

The first set of lines in the **license.xml** file, starting with the RMS license file version (15), all have the same locking code starting with *.

The following example code comes just after the NON_REDUNDANT text.

```
15 PACK_ERM-CS 1.0 LONG NORMAL NETWORK AGGR 2_KEYS INFINITE_KEYS 8 JUL 2015
1 DEC 2015 APP_SHR:254 SLM_CODE 2_LOG_ENCRYPTION_LEVEL NON_COMMUTER
NO_GRACE NO_OVERDRAFT FLOAT NON_REDUNDANT *15QL6UFV6WTR5FW2014,Ni NiL Ni
```

In this example it is *15QL6UFV6WTR5FW2014

7.7 How can I check the machine is in the same state as the license code?

Run the program: **C:\AVEVA\AVEVA Licensing System\RMS\Tools\Wechoid.exe**

Click to select the **Disk ID**, **Ethernet Address** and **UUID** check boxes.

In the example, the Code plus the Selector Values added together make up the locking code in the **license.xml** file as above.

The screenshot shows the 'Wechoid' dialog box with the following fields and values:

Locking Criteria	Value
☐ IP Address	172.31.200.68
☐ ID PROM	
☐ Disk ID	0x0C9F5280
☐ Host Name	DCS010951
☒ Ethernet Address	68-94-23-00-2F-72
☐ Computer ID	
☐ Hard Disk Serial	WD-WXF1C52M3753
☐ Standard Custom	
☐ CPU Info String	GenuineIntel Intel(R) Core(TM) i7-2760QM
☒ UUID	4C4C4544-004D-3610-8034-B9C04F425431
☐ Extended Custom	

Below the criteria, the 'Locking Data' section shows:

- ☒ New Style (selected) and ☐ Old Style
- Selector: 0x2010
- Code: *1CANZQMD523 CEPZ

An 'OK' button is at the bottom.

Locking code = *1CANZQMD523CEPZ plus the last part of the selector 0x2010 = *1CANZQMD523CEPZ2010

7.8 How to check that the output from echoid.exe is the same as from running Wechoid.exe?

Run the C:\AVEVA\AVEVA Licensing System\RMS\Tools\AVEVA-locking.bat file.

Check that the output from **echoid.exe** is the same as from running **Wechoid.exe**.

The output from echoid (Disk ID, Ethernet address[1], and UUID) is the same as the output from **Wechoid.exe** above.

Sentinel RMS Development Kit 8.5.3.0020 Host Locking Code Information Utility Copyright © 2012 SafeNet, Inc.

IP address	: 10.0.150.17
Disk ID	: 0x884BA25A
Host name	: UKCAMD3903
Ethernet address[1]	: 18-A9-05-C2-37-21
Hard Disk Serial[1]	: 9VMC5AEK

CPU Info String : GenuineIntel Intel(R) Xeon(R) CPU W3520 @ 2.67GHz
6 10 5

UUID : BD5E286E-30D0-11DF-BBDA-05C2372118A9

If different values are obtained at the time of installing the license, there may be a problem; contact Support/License development.

If locking errors still occur, send the following information to the AVEVA License Administrator:

- The License file
- The AVEVA-Server-UKCAMD3903-info.txt file
- The screen shot of running wechoid.exe as shown above.

7.9 What should I do if I get the error 'locking code mismatch' during the license file installation even though I specified the correct values?

This usually happens on laptop installs. The license server state has changed since requesting the license servers locking criteria, and now when attempting to start the license server you get a locking-code mismatch.

For example, when running the **AVEVA-Locking.bat** file, the laptop was using the LAN network card, and now, when attempting to start the license server, the laptop is using the wireless network adaptor.

For example, the following are the contents of a **AVEVA-Server-info.txt** file when a server is connected over the LAN with wireless network card disabled.

IP address : 10.0.101.38
Disk ID : 0x8771CA45
Host name : UKCAML4161
Ethernet address [1] : 2C-41-38-08-01-9F
Hard Disk Serial [1] : J3360081G1KVBC
CPU Info String : GenuineIntel Intel(R) Core(TM) i5-2540M CPU @
2.60GHz 6 10 7
UUID : 008E337B-321F-11E1-9716-6CC25F06A060

Same machine with Wireless network active (see two Ethernet addresses now):

IP address : 10.0.101.38
Disk ID : 0x8771CA45
Host name : UKCAML4161
Ethernet address [1] : AC-81-12-97-F1-1C
Hard Disk Serial [1] : J3360081G1KVBC

```

CPU Info String      : GenuineIntel Intel(R) Core(TM) i5-2540M CPU @
                        2.60GHz 6 10 7

UUID                : 008E337B-321F-11E1-9716-6CC25F06A060

Locking Code 1 [1]   : 381E-*15S SFAW CGQ8 XADQ

Ethernet address [2] : 2C-41-38-08-01-9F

Hard Disk Serial [1] : J3360081G1KVBC
    
```

Two workarounds are available:

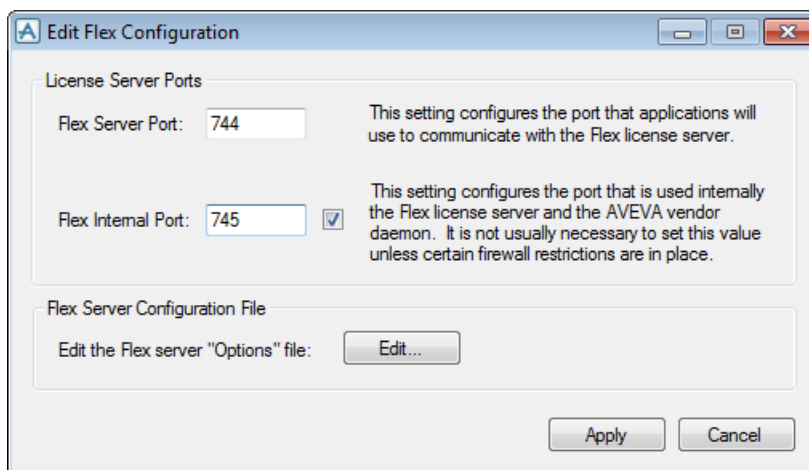
Place the license server into the same state (as when the **.bat** file was run when the license was requested), then run the license.

Note: Once the license file has been installed, it will work, which ever network card is being used, so long as the original card is active /enabled.

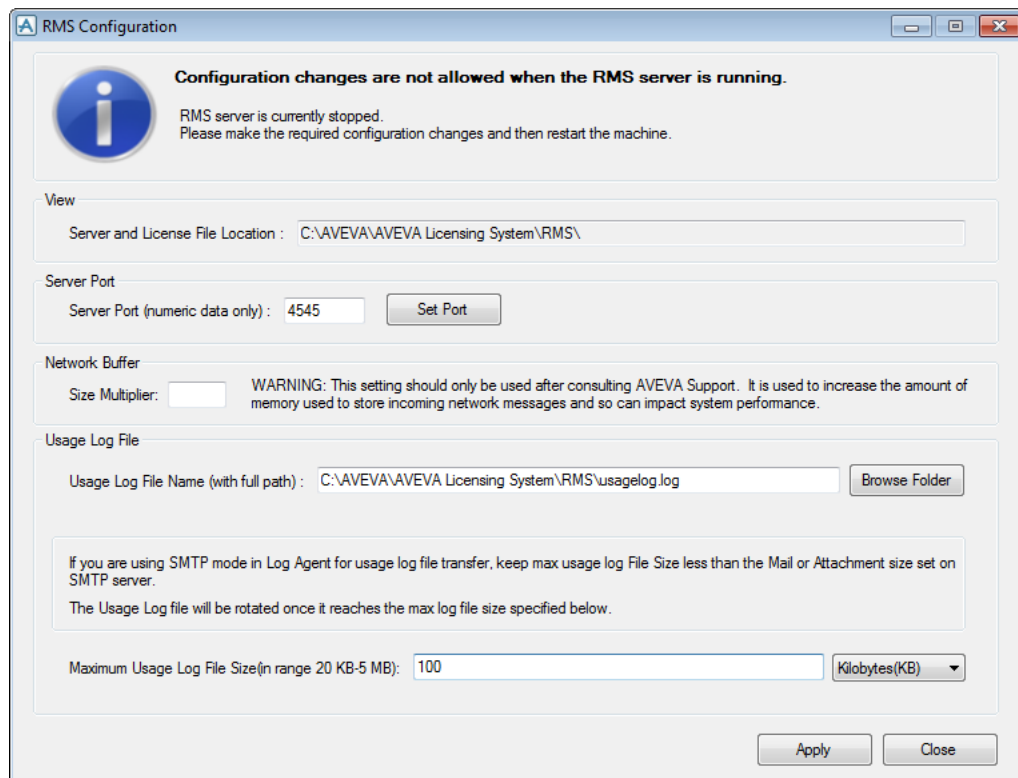
Run the (**.bat** file) again and request a new license file with the new [1] locking criteria. Once obtained, install it without changing the server state.

7.10 How do I specify all TCP and UDP ports for ALS through a firewall?

Flex server ports can be set via the **AVEVA License System Console > Flex - Edit configuration** in any ALS1.1.1 version and later versions.

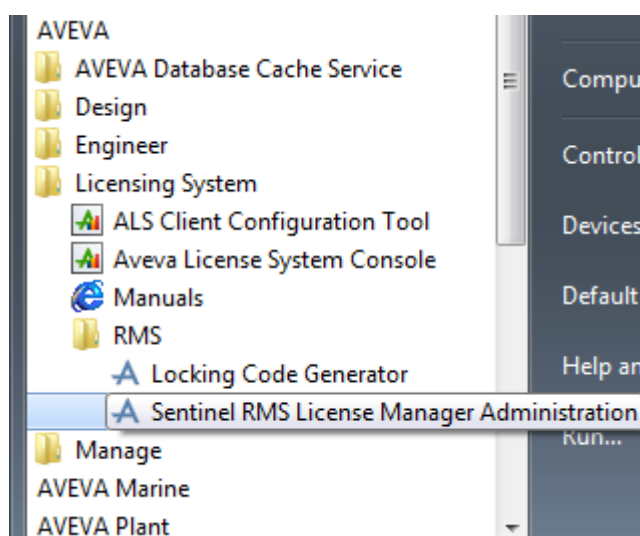


The RMS server port can be set via the **ALS console > RMS - Edit configuration** in any ALS version.



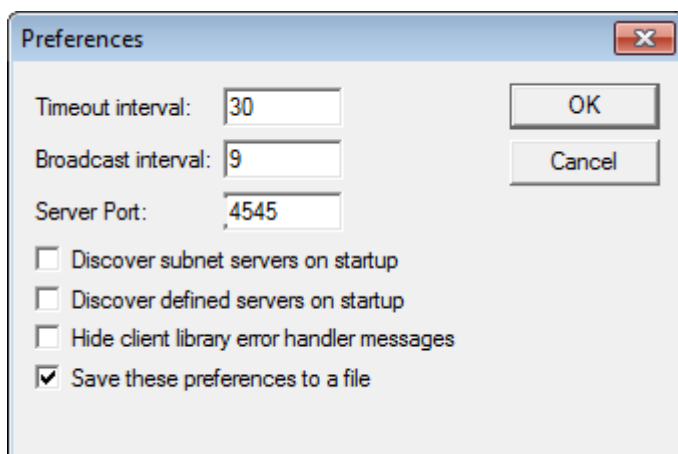
7.11 How can I see users accessing my ALS/RMS license server?

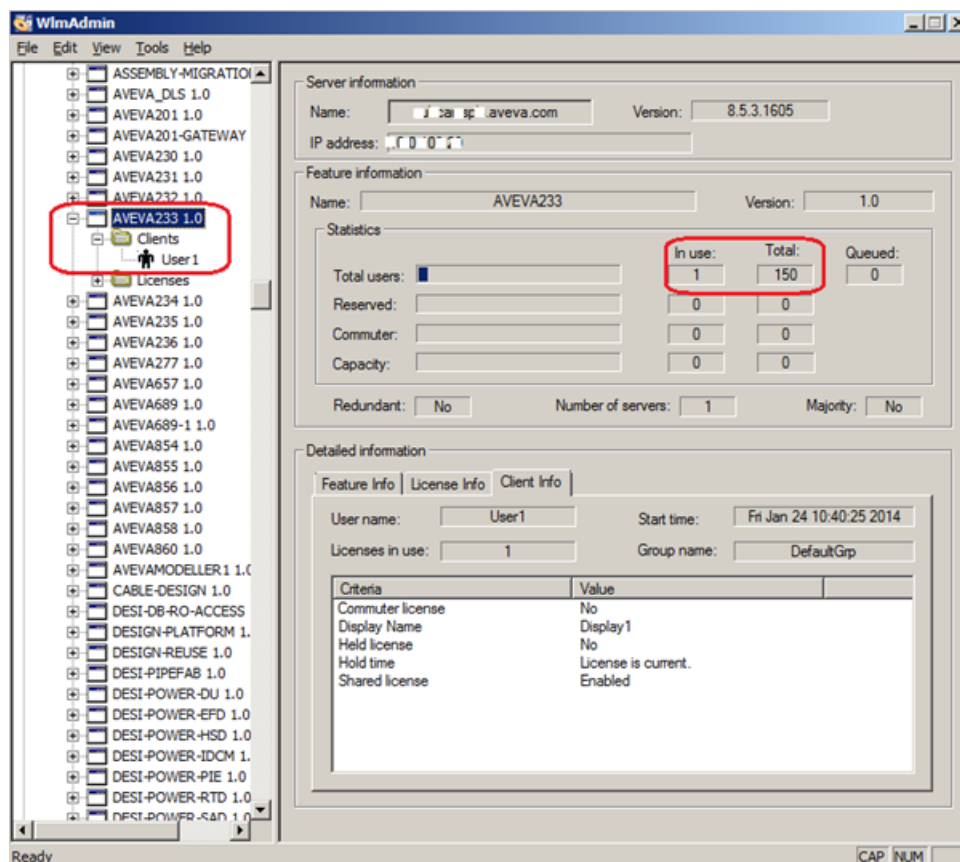
All product usage for both Flex-secured products and RMS-secured products is displayed using the Sentinel RMS License Manager Administration tool.



By default, the tool looks for RMS license servers on the default RMS port of 5093. However AVEVA's default RMS port is 4545. To correct this:

1. Select **Edit > Preferences** and enter the server port as 4545, then click to select the **Save these preferences** check box.
2. Close and start the **Sentinel RMS License Manager Administration** tool.
3. Expand the Subnet servers and find the required server name.
4. Click the license feature.
5. If the server is not listed under subnet servers then select **Edit > Defined Server List**.
6. Enter the server name, then click **Add** then **OK**.





7.12 What should I do if FLEXlm lmstat-a does not show new RMS secured product usage?

FLEXlm lmstat-a does not show new RMS secured product usage. All the product usage for old Flex secured products and new RMS secured products are shown in the RMS server. It is the RMS server which controls product usage NOT Flex.

Flex/Lmstat -a will only show clients accessing FlexLM secured products.

To see ALL of the AVEVA product usage, use the **Sentinel RMS License Manager Administration** tool that is supplied with ALS.

Or

Use the **lsmon.exe** to view the details of all usage. It can be found in the path **C:\AVEVA\AVEVA Licensing System\RMS\Tools** folder. Create a **.bat** file in the same folder with the following lines in it: -

```
cd c:\AVEVA\AVEVA Licensing system\RMS\Tools
lsmon.exe yourservername >license-used.txt
```

Change yourservername to your server name and run the .bat file. It will create a file called **license-used.txt** with a snapshot of the keys in the license file and the clients accessing them.

7.13 How do I perform a silent installation of the ALS Client configuration Tool?

To silently roll out the ALS Client configuration tool to all users:

The ALS installer .msi can be run by command line to silently install the whole product by using the command which is supplied along with a silent-30363(release number).bat file if the ALS release is on a DVD release.

The bat file basically runs msixec sending the output to a logfile named with the date and time, as follows:

```
set mndate=%DATE:/=-%
```

```
set mntime=%TIME:/=-%
```

```
set
```

```
logfile=C:\AVEVA\sinstallog\AVEVA_Licensing_install_%mndate%-%mntime%.log
```

Then msixec is run to install ALL ALS writing the output to the logfile.

Example:

```
msiexec /i "ALS103\Licensing1.03.msi" /qn REBOOT="ReallySuppress" /l*e "%logfile%" /passive ADDLOCAL=ALL
```

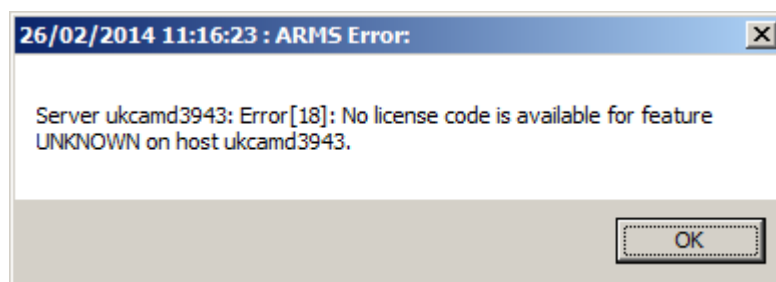
Now to install the Client change ADDLOCAL=ALL to ADDLOCAL=CLI,CLIHLP. If they do not want the Client Help files then it can be reduced to ADDLOCAL=CLI

Example:

```
msiexec /i "ALS103\Licensing1.03.msi" /qn REBOOT="ReallySuppress" /l*e "%logfile%" /passive ADDLOCAL=CLI
```

7.14 What should I do if I get Error [18] when I run the RMS license secured products (ex. Aveva E3D)?

When running any product for the first time, the following error message is displayed:



There are several possible reasons for this.

- This could mean that the Specific license key that the product is trying to use is NOT installed in the RMS license server which is running. A common reason is running AVEVA ADMIN with a license file with just an AVEVA E3D seat and not one with AVEVA ADMIN
- This could also be due to the new license file being linked to either the wrong locking criteria or a mistake has been made when copying and pasting

- The license file in the server has not been installed yet

Run the AVEVA-Locking.bat file in the c:\AVEVA\AVEVA Licensing System\RMS\Tools directory.

Send the resulting (**AVEVA-Server-SERVERNAME-info.txt**), along with the **license.xml** file back to the Administrator who issued the license, with a screen shot of the product/ version that produces the error.

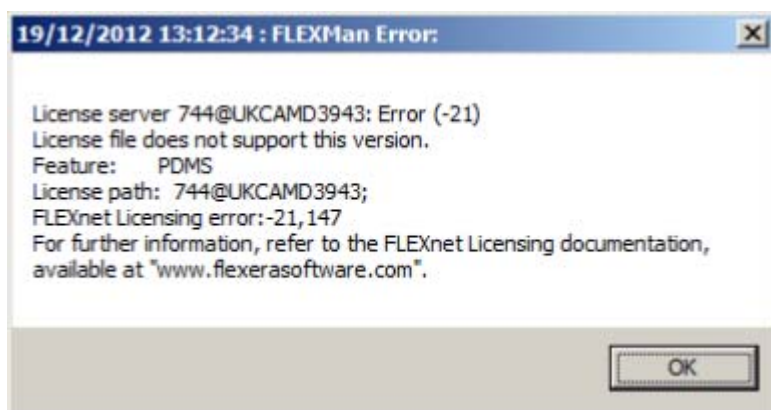
7.15 Why am I getting Error 8 in the Flex license secured products?

If a new Flex 6 licenses.lic part from an installed AVEVA Licensing System server is placed into an old Flexman5.2 server then following error message is displayed. The example is from a PDMS application.



7.16 Why am I getting Error -21 in the Flex license secured products?

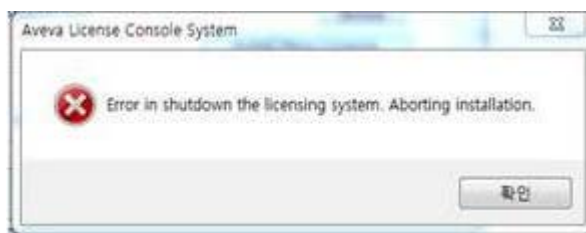
If an old Flex 5 license file is placed into the Flex part of the AVEVA Licensing System server, the following error message is displayed. This following example is from running a PDMS application.



7.17 What should I do if get the error 'Aborting Installation' when I try to install the license file?

If all the Flex license keys have expired on a license server, the **cadcentre.exe** process aborts but the main Flex license server process (**lmgrd.exe**) is still running.

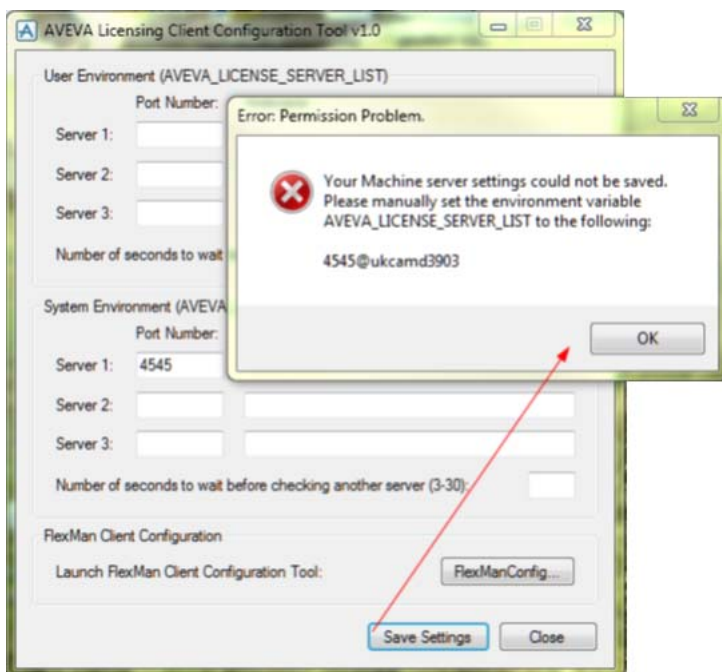
When an attempt to install a new license file on that server occurs, the AVEVA license System Console thinks that the servers are down, but in fact the Flex server/service is still running. When this happens the following error is displayed.



To overcome this error, the Flex service (**lmgrd.exe**) programs must be stopped, this can be done by stopping the **lmgrd.exe** (Aveva DLS Flex service) processes. Once the service is stopped the new license can be installed.

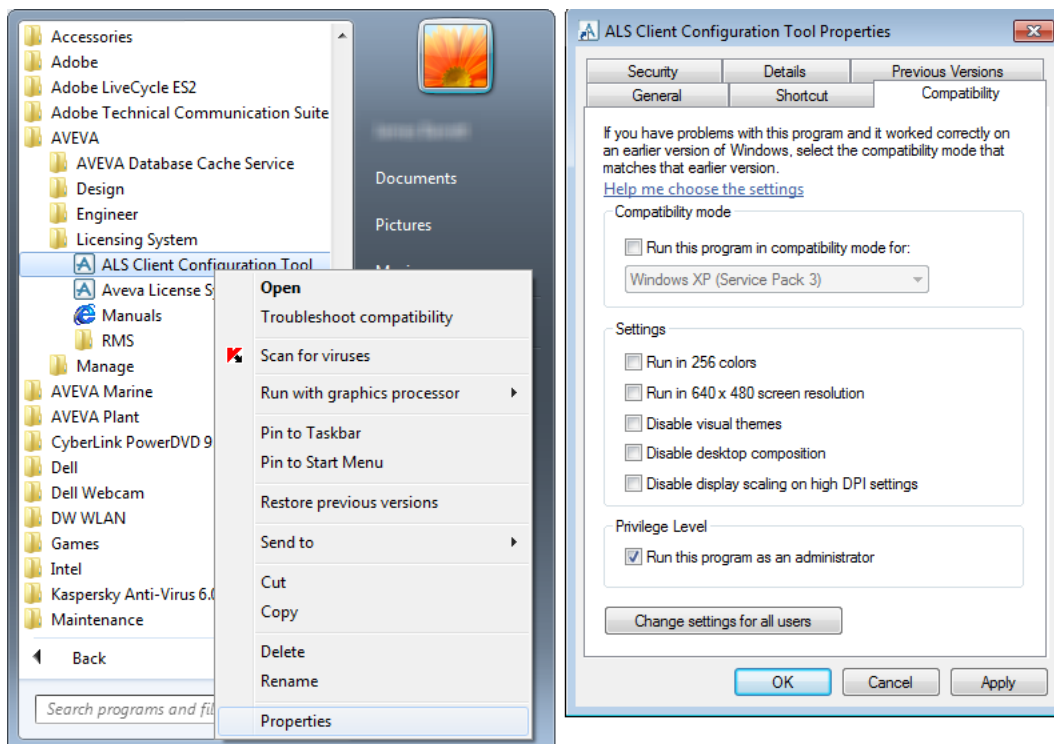
7.18 What should I do if I get the error 'Permission Problem' when attempting to set the port@hostname in the ALS client config tool?

When setting up a client to use an AVEVA licensing server, the following error is displayed when running the ALS Client Configuration Tool.



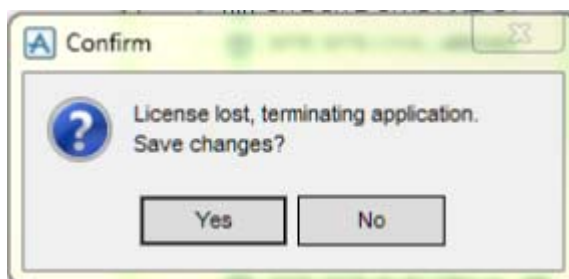
This means that the program must be run as an administrator.

Select the ALSClientConfigTool.exe, right click > **Properties**. Select the **Compatibility** tab and set the privilege Level to, **Run this program as an administrator**.



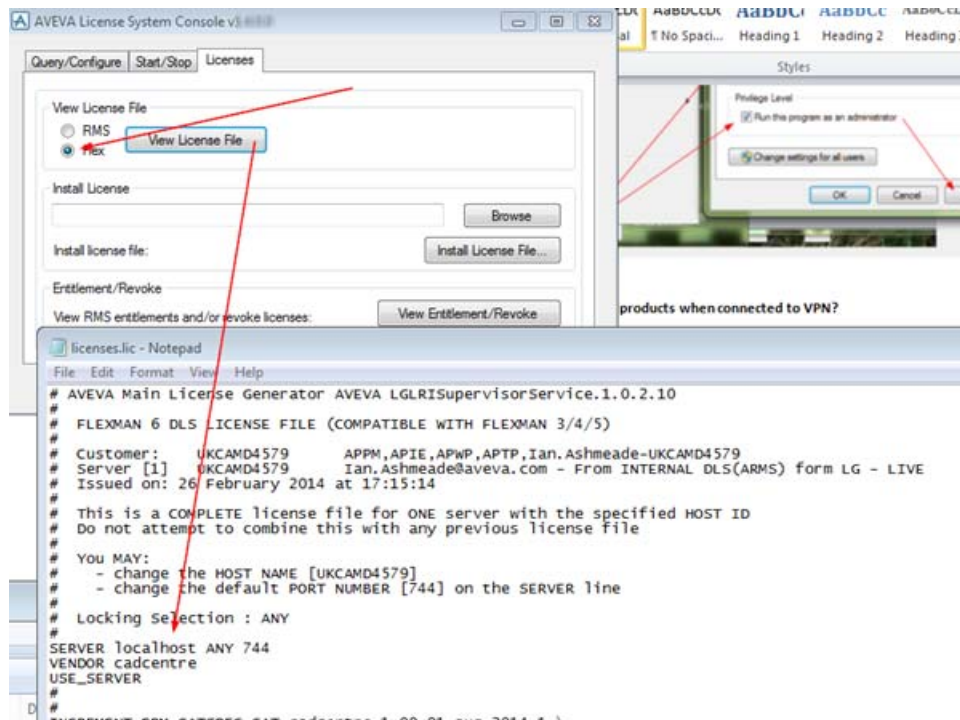
7.19 Why am I having issues when running the products whilst connected to VPN?

If using a local AVEVA Licensing server and a VPN client, when stopping the VPN client the license server stops, or the route to the license server is not restored, and after 8-10 minutes the applications return server lost prompts.



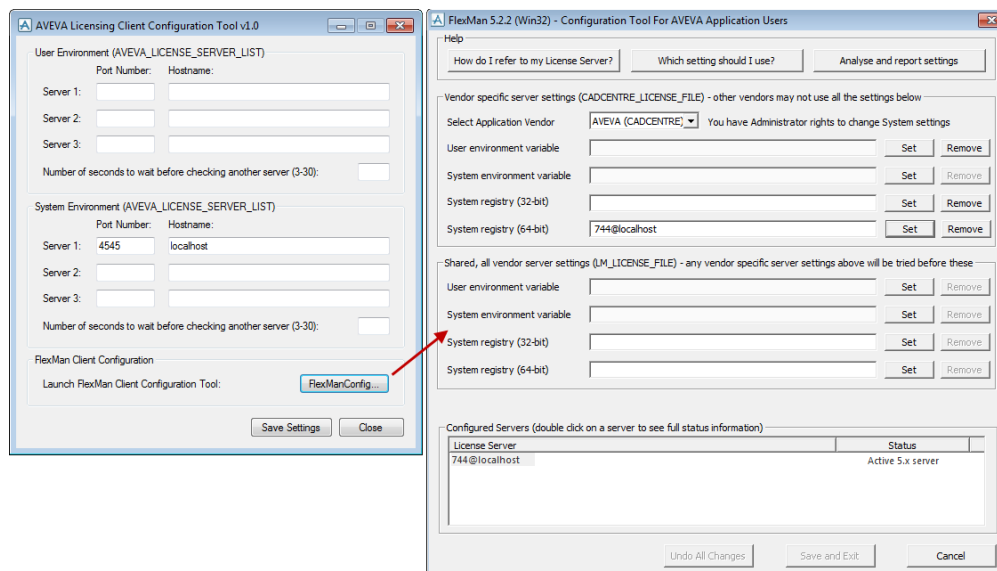
The only way to continue is to restart the license server or re-boot the machine. This can be fixed if both the RMS and Flex parts of the AVEVA Licensing System are pointed to the localhost.

This can be done by editing the **Flex licenses.lic** file using the AVEVA License System Console.

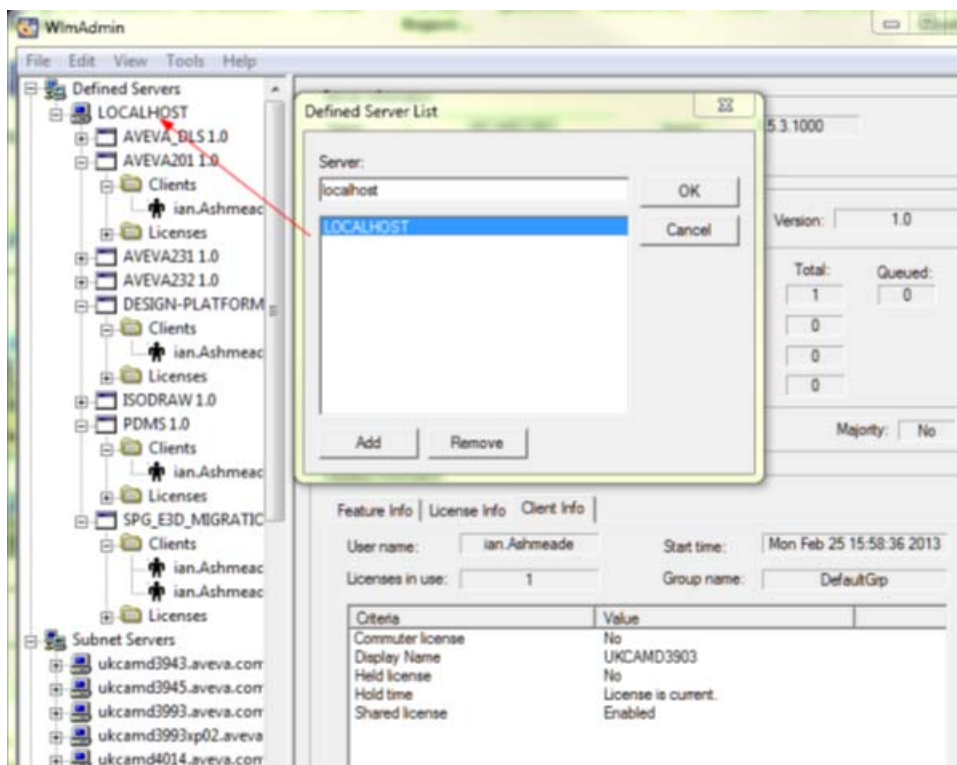


Change the SERVER line to read: SERVER localhost ANY 744.

This must be done manually each time a new license is installed. Similarly, the ALS client config must be changed to look for both the RMS and Flex servers on localhost:



Use the **Sentinel RMS License Manager Administration** tool to check on the license server usage after configuring localhost as above, add a Defined Server for localhost..

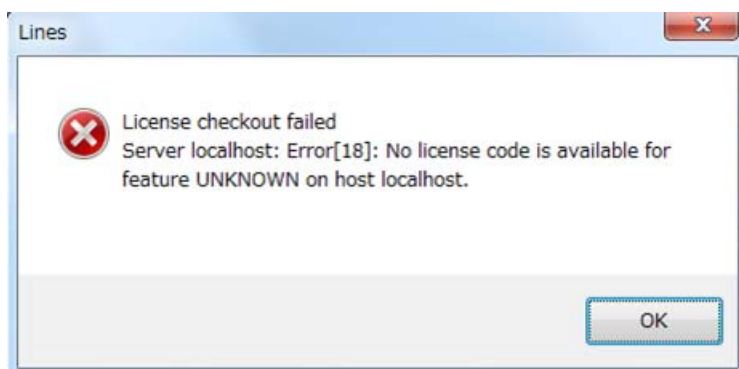


7.20 Why am I getting Error [18] and Error (-602) after revoking the licenses?

This error occurs when the products are run five days after revoking part of the licences or all licenses.

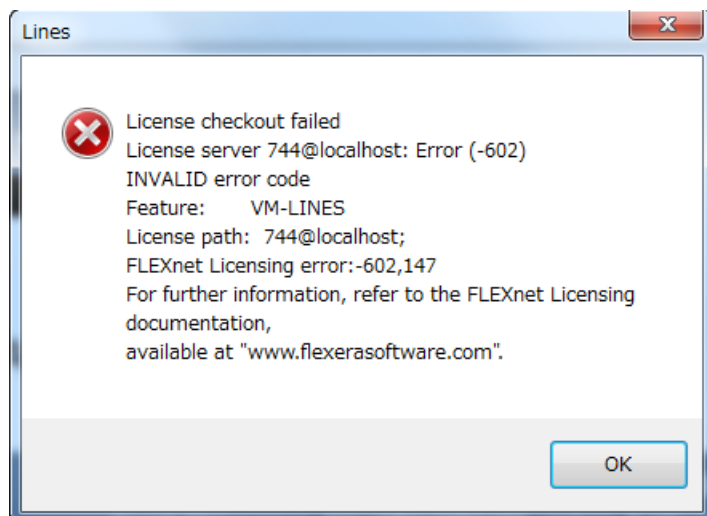
When running a RMS secured product (ex. AVEVA E3D) the following error is displayed:

Error[18]: No license code is available for feature UNKNOWN on host localhost.



When running an old FLEX secured product (ex. PDMS) the following error is displayed:

Error (-602) INVALID error code



The license server appears to be up and running, but once the file **RevokeRequest.xml** is created and placed in the folder: **C:\AVEVA\AVEVA Licensing System\RMS**, any products will cease to run after five days.

7.21 List of Error Codes -601 to -610 displayed by the Flex clients

The new DLS is not able to convey a meaningful error code back to the Flex based applications due to the way that the Flex vendor daemon framework works and the fact that RMS error codes do not align with Flex error codes.

-601: Unable to create a FLUID.

This indicates a problem with the server system environment or possibly a problem with the DLS server implementation.

-602: Can't get RMS feature info.

This indicates that either the RMS server is offline or some other problem with the server.

-603: Feature renewal time is below the allowed minimum.

This indicates that the RMS license has been incorrectly generated.

-604: Unable to set sharing attributes.

This indicates a problem with the RMS server.

-605: Can't set Customer Data field. (HOST USERNAME DISPLAY).

This indicates a problem with the RMS server.

-606: RMS AF Checkout failed.

Either the licenses have been used, the RMS server is down, or some other problem.

-607: RMS PF Checkout failed.

Either the licenses have been used; the RMS server is down, a problem with the way the license was generated, or some other problem.

-608: Failed to create map structure.

System problem.

-609: Failed to store map structure.

System problem.

-610: Failed to store map structure.

System problem.



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